Section 4.1 Solutions

Study Guide

February 28, 2026

Problem 1

We are asked to find integers q and r such that $n = dq + r$ and $0 \leq r < d$, given $n = 70$ and $d = 9$.

We need to find the quotient q and the remainder r when 70 is divided by 9. We can do this by performing division:

$70 \div 9$

We look for the largest multiple of 9 that is less than or equal to 70. The multiples of 9 are $9, 18, 27, 36, 45, 54, 63, 72, \ldots$. The largest multiple of 9 that is less than or equal to 70 is 63.

We can express 70 as a multiple of 9 plus a remainder: $70 = 9 \times q + r$

Since $63 = 9 \times 7$, we have: $70 = 9 \times 7 + r$

To find the remainder r, we subtract 63 from 70: $r = 70 - 63$ $r = 7$

So, we have $q = 7$ and $r = 7$. We check if the condition $0 \leq r < d$ is satisfied. Here, $d = 9$. So, we check if $0 \leq 7 < 9$. This inequality is true.

Therefore, the integers q and r are $q = 7$ and $r = 7$.

We can write the equation as: $70 = 9 \times 7 + 7$

The final answer is $\boxed{q = 7, r = 7}$.

Problem 2

2. If n and d are integers with d ¿ 0, n div d is q and n mod d is r.

According to the quotient-remainder theorem, for any integers n and d with $d > 0$, there exist unique integers q and r such that $n = dq + r$ and $0 \leq r < d$.

The term 'n div d' represents the quotient q when n is divided by d. The term 'n mod d' represents the remainder r when n is divided by d.

Therefore, if n and d are integers with $d > 0$, n div d is q and n mod d is r.

Problem 3

Problem 3: The parity of an integer indicates whether the integer is $_{or}$.

The parity of an integer indicates whether the integer is **even** or **odd**.

Problem 4

The problem asks to find integers q and r such that $n = dq + r$ and $0 \leq r < d$ for $n = 3$ and $d = 11$.

We are given $n = 3$ and $d = 11$. We need to find integers q and r such that $3 = 11q + r$ and $0 \leq r < 11$.

We can observe that if we choose $q = 0$, then the equation becomes:

$$3 = 11(0) + r$$

$$3 = 0 + r$$

$$r = 3$$

Now we check if the condition $0 \leq r < d$ is satisfied. We have $r = 3$ and $d = 11$. Is $0 \leq 3 < 11$? Yes, this condition is satisfied.

Therefore, for $n = 3$ and $d = 11$, we have $q = 0$ and $r = 3$.

The final answer is $\boxed{q = 0, r = 3}$.

Problem 5

The problem asks us to find integers q and r such that $n = dq + r$ and $0 \leq r < d$, given $n = 3$ and $d = 11$.

We are given $n = 3$ and $d = 11$. We need to find integers q and r such that $n = dq + r$ and $0 \leq r < d$. Substituting the given values, we have $3 = 11q + r$ and $0 \leq r < 11$.

We can choose $q = 0$. Then, the equation becomes $3 = 11(0) + r$. This simplifies to $3 = 0 + r$, which means $r = 3$.

Now, we check if the condition $0 \leq r < d$ is satisfied. We have $r = 3$ and $d = 11$. Is $0 \leq 3 < 11$? Yes, this inequality is true.

Therefore, the integers q and r are $q = 0$ and $r = 3$.

We can write this as: $n = dq + r$ $3 = 11 \times 0 + 3$

The final answer is $\boxed{q = 0, r = 3}$.

Problem 6

The problem asks us to find integers q and r such that $n = dq + r$ and $0 \leq r < d$, given $n = -27$ and $d = 8$.

We are given $n = -27$ and $d = 8$. We want to find integers q and r such that $-27 = 8q + r$ and $0 \leq r < 8$.

We can start by finding a multiple of $d = 8$ that is close to $n = -27$. Let's consider some multiples of 8: $8 \times 0 = 0$ $8 \times (-1) = -8$ $8 \times (-2) = -16$ $8 \times (-3) = -24$ $8 \times (-4) = -32$

We want to find a q such that $8q$ is close to -27 and when we calculate $r = -27 - 8q$, the condition $0 \leq r < 8$ is satisfied.

Let's try $q = -3$. If $q = -3$, then $8q = 8 \times (-3) = -24$. Now, we calculate $r = n - 8q$. $r = -27 - (-24)$ $r = -27 + 24$ $r = -3$ This value of r does not satisfy the condition $0 \leq r < 8$ because -3 is not in the range $[0, 8)$.

Let's try $q = -4$. If $q = -4$, then $8q = 8 \times (-4) = -32$. Now, we calculate $r = n - 8q$. $r = -27 - (-32)$ $r = -27 + 32$ $r = 5$ This value of r satisfies the condition $0 \leq r < 8$ because $0 \leq 5 < 8$.

So, we have found integers $q = -4$ and $r = 5$ such that $n = dq + r$ and $0 \leq r < d$. We can check this: $-27 = 8 \times (-4) + 5$ $-27 = -32 + 5$ $-27 = -27$ The equation holds true. The remainder $r = 5$ satisfies the condition $0 \leq 5 < 8$.

Therefore, the integers q and r are $q = -4$ and $r = 5$.

The final answer is $\boxed{q = -4, r = 5}$.

Problem 7

Here's the solution to problem 7:

7. Evaluate the expressions in 7-10. a. 43 div 9 b. 43 mod 9

To evaluate these expressions, we use the Quotient-Remainder Theorem, which states that for any integers n and d with $d > 0$, there exist unique integers q and r such that $n = dq + r$ and $0 \leq r < d$. Here, q is the quotient and r is the remainder.

Part a: 43 div 9

We need to find the quotient when 43 is divided by 9. We look for an integer q such that $9q \leq 43$.

$$
\begin{aligned}
9 \times 1 &= 9 \\
9 \times 2 &= 18 \\
9 \times 3 &= 27 \\
9 \times 4 &= 36 \\
9 \times 5 &= 45
\end{aligned}
$$

Since $9 \times 4 = 36$ is the largest multiple of 9 that is less than or equal to 43, the quotient is 4.

Therefore, 43 div 9 = 4.

Part b: 43 mod 9

Now we need to find the remainder when 43 is divided by 9. We use the equation from the Quotient-Remainder Theorem: $n = dq + r$. We have $n = 43$, $d = 9$, and we found $q = 4$.

$$
\begin{aligned}
43 &= 9 \times 4 + r \\
43 &= 36 + r \\
43 - 36 &= r \\
7 &= r
\end{aligned}
$$

The remainder is 7. We check if the condition $0 \leq r < d$ is satisfied: $0 \leq 7 < 9$. This is true.

Therefore, 43 mod 9 = 7.

Summary: a. 43 div 9 = 4 b. 43 mod 9 = 7

Problem 8

Here's the solution to problem 8:

We need to evaluate the expressions 50 div 7 and 50 mod 7.

8. a. 50 div 7

We are looking for integers q and r such that $50 = 7q + r$, where $0 \leq r < 7$.

We can perform the division of 50 by 7.

$$50 \div 7 = 7 \text{ with a remainder of } 1$$

So, we can write:

$$50 = 7 \times 7 + 1$$

Here, $q = 7$ and $r = 1$. Since $0 \leq 1 < 7$, this is the correct representation.

Therefore, 50 div 7 is the quotient, which is 7.

$$50 \text{ div } 7 = 7$$

8. b. 50 mod 7

We are looking for the remainder when 50 is divided by 7. From the division performed in part (a), we found that:

$$50 = 7 \times 7 + 1$$

The remainder is the value of r in this equation, where $0 \leq r < 7$. In this case, $r = 1$.

Therefore, 50 mod 7 is the remainder, which is 1.

$$50 \text{ mod } 7 = 1$$

Problem 9

Here's the solution for problem 9:

We are asked to evaluate the expressions 28 div 5 and 28 mod 5.

We need to find integers q and r such that $n = dq + r$ and $0 \leq r < d$, where $n = 28$ and $d = 5$.

$$28 = 5 \cdot q + r$$

We need to find the largest multiple of 5 that is less than or equal to 28. The multiples of 5 are: 5, 10, 15, 20, 25, 30, ...

The largest multiple of 5 less than or equal to 28 is 25.

$$25 = 5 \cdot 5$$

So, we can write 28 as:

$$28 = 25 + 3$$
$$28 = 5 \cdot 5 + 3$$

In this case, $q = 5$ and $r = 3$. We check if the condition $0 \leq r < d$ is satisfied. $0 \leq 3 < 5$, which is true.

Therefore, 28 div $5 = q = 5$ 28 mod $5 = r = 3$

The final answer is $\boxed{a.5, b.3}$.

Problem 10

To evaluate the expressions in problem 10, we need to find the quotient and remainder when 30 is divided by 2.

The quotient-remainder theorem states that for any integer n and any positive integer d, there exist unique integers q and r such that $n = dq + r$ and $0 \leq r < d$.

In this case, $n = 30$ and $d = 2$. We are looking for integers q and r such that $30 = 2q + r$ and $0 \leq r < 2$.

We can perform the division of 30 by 2.

$$\frac{30}{2} = 15$$

So, the quotient q is 15.

Now, we can find the remainder r. We can use the equation $n = dq + r$ and substitute the values of n, d, and q.

$$30 = 2(15) + r$$

$$30 = 30 + r$$

Subtracting 30 from both sides of the equation:

$$30 - 30 = r$$

$$0 = r$$

The remainder r is 0.

We need to check if the condition $0 \leq r < d$ is satisfied. Here, $r = 0$ and $d = 2$. So, $0 \leq 0 < 2$, which is true.

Therefore, for $n = 30$ and $d = 2$: a. 30 div 2 is the quotient, which is $q = 15$. b. 30 mod 2 is the remainder, which is $r = 0$.

The final answer is $\boxed{\text{a. 15, b. 0}}$.

Problem 11

Here's the solution to problem 11, showing each algebraic step clearly:

11. Check the correctness of formula (4.4.1) given in Example 4.4.3 for the following values of DayT and N.

Formula (4.4.1) is given by:

$$\text{Day}(N) = (\text{DayT} + N) \mod 7$$

a. DayT = 6 (Saturday) and N = 15

First, we calculate (DayT + N):

$$\text{DayT} + N = 6 + 15$$

$$\text{DayT} + N = 21$$

Now, we apply the modulo 7 operation:

$$\text{Day}(15) = 21 \mod 7$$

$$\text{Day}(15) = 0$$

According to the convention where Sunday is 0, a result of 0 corresponds to Sunday.

b. DayT = 0 (Sunday) and N = 7

First, we calculate (DayT + N):

$$\text{DayT} + N = 0 + 7$$

$$\text{DayT} + N = 7$$

Now, we apply the modulo 7 operation:

$$\text{Day}(7) = 7 \mod 7$$

$$\text{Day}(7) = 0$$

A result of 0 corresponds to Sunday.

c. DayT = 4 (Thursday) and N = 12

First, we calculate (DayT + N):

$$\text{DayT} + N = 4 + 12$$

$$\text{DayT} + N = 16$$

Now, we apply the modulo 7 operation:

$$\text{Day}(12) = 16 \mod 7$$

To find 16 mod 7, we divide 16 by 7:

$$16 = 2 \times 7 + 2$$

The remainder is 2. So,

$$\text{Day}(12) = 2$$

A result of 2 corresponds to Tuesday.

Problem 12

12. Justify formula (4.4.1) for general values of DayT and N.

The problem refers to formula (4.4.1) from Example 4.4.3, which is not provided in the text. However, based on the context of the preceding questions involving days of the week and the quotient-remainder theorem, formula (4.4.1) likely relates to calculating the day of the week after a certain number of days have passed, given a starting day. A common formula for this type of problem is:

Day of the week = (Starting Day + Number of Days) mod 7

Here, we can represent the days of the week with numbers: Sunday = 0 Monday = 1 Tuesday = 2 Wednesday = 3 Thursday = 4 Friday = 5 Saturday = 6

Let D_{start} be the numerical representation of the starting day of the week. Let N be the number of days that have passed. The day of the week after N days can be calculated as $(D_{start} + N) \pmod 7$.

However, the question mentions "DayT" and "N". Looking at question 11, the formula seems to be applied as: DayT = 6 (Saturday) and N = 15. The result is checked against some formula. DayT = 0 (Sunday) and N = 7. The result is checked against some formula. DayT = 4 (Thursday) and N = 12. The result is checked against some formula.

This suggests that $DayT$ might be the numerical representation of the starting day of the week, and N is the number of days. The formula (4.4.1) likely is:

Resulting Day = (DayT + N) mod 7

We need to justify this formula for general values of DayT and N, where $DayT \in \{0, 1, 2, 3, 4, 5, 6\}$ and N is a non-negative integer.

Justification:

The days of the week repeat in a cycle of 7. This means that after every 7 days, the day of the week is the same. This cyclical nature is precisely what the modulo 7 operation captures.

Let's assume $DayT$ is the numerical value representing the starting day of the week, where $DayT \in \{0, 1, 2, 3, 4, 5, 6\}$. Let N be the number of days that have passed.

We want to find the day of the week after N days. If N is a multiple of 7, say $N = 7k$ for some integer k, then after N days, the day of the week will be the same as the starting day. In this case, $(DayT + N) \pmod 7 = (DayT + 7k) \pmod 7$. Since $7k$ is a multiple of 7, $7k \pmod 7 = 0$. Therefore, $(DayT + 7k) \pmod 7 = DayT \pmod 7 = DayT$. This is consistent with the fact that after a whole number of weeks, we return to the same day.

If N is not a multiple of 7, we can use the quotient-remainder theorem to write N in the form $N = 7q + r$, where q is the quotient and r is the remainder, with $0 \leq r < 7$. Here, q represents the number of full weeks that have passed, and r represents the remaining days.

After q full weeks, the day of the week will be the same as the starting day, $DayT$. The remaining r days will then advance the day of the week. So, the resulting day of the week will be equivalent to starting from $DayT$ and advancing by r days.

Therefore, the resulting day of the week can be calculated as:

$$\text{Resulting Day} \equiv DayT + r \pmod 7$$

Substituting $N = 7q + r$ into the expression $(DayT + N) \pmod 7$:

$$(DayT + N) \pmod 7 = (DayT + (7q + r)) \pmod 7$$

$$(DayT + N) \pmod 7 = (DayT + 7q + r) \pmod 7$$

Since $7q$ is a multiple of 7, $7q \pmod 7 = 0$.

$$(DayT + 7q + r) \pmod 7 = (DayT + r) \pmod 7$$

This shows that the formula $(DayT + N) \pmod 7$ correctly accounts for the cyclical nature of the days of the week. The modulo 7 operation ensures that the result always falls within the range of possible day representations (0 to 6).

Thus, the formula (4.4.1), which is likely of the form $(DayT + N) \pmod 7$, is justified for general values of $DayT$ and N because it correctly models the 7-day cycle of the week.

Problem 13

To determine the day of the week after 30 days, we can use the concept of modular arithmetic, as there are 7 days in a week.

Let Monday be represented by the number 1. The days of the week can be represented as follows: Sunday = 0 Monday = 1 Tuesday = 2 Wednesday = 3 Thursday = 4 Friday = 5 Saturday = 6

The problem states that today is Monday. We need to find the day of the week after 30 days.

We can find the day of the week by calculating: (Today's day number + Number of days) mod 7

Today's day number for Monday is 1. The number of days is 30.

So, we need to calculate: $(1 + 30) \mod 7$

First, sum the numbers: $1 + 30 = 31$

Now, find the remainder when 31 is divided by 7: $31 \div 7$

We can perform the division: $31 = 7 \times 4 + 3$

The quotient is 4 and the remainder is 3. So, $31 \mod 7 = 3$.

The remainder 3 corresponds to Wednesday.

Alternatively, we can first find the remainder of the number of days when divided by 7, and then add it to today's day number.

Number of days = 30 Calculate $30 \mod 7$: $30 = 7 \times 4 + 2$ So, $30 \mod 7 = 2$.

This means that after 30 days, it will be the same day of the week as 2 days from Monday.

Today is Monday. 1 day from Monday is Tuesday. 2 days from Monday is Wednesday.

Both methods yield the same result.

The day of the week after 30 days will be Wednesday.

The final answer is $\boxed{Wednesday}$.

Problem 14

Here's the solution to problem 14:

Problem 14: On a Monday a friend says he will meet you again in 30 days. What day of the week will that be?

Solution:

We can represent the days of the week with integers. Let Sunday be 0, Monday be 1, Tuesday be 2, Wednesday be 3, Thursday be 4, Friday be 5, and Saturday be 6.

[10pt]

We are given that today is Monday. So, the current day of the week can be represented by 1.

[10pt]

We want to find the day of the week after 30 days. We need to find the remainder when 30 is divided by 7 (the number of days in a week).

[10pt]

We can use the division algorithm, which states that for any integer n and any positive integer d, there exist unique integers q (quotient) and r (remainder) such that $n = dq + r$ and $0 \leq r < d$. In this case, $n = 30$ (the number of days) and $d = 7$ (the number of days in a week).

[10pt]

We want to find q and r such that:

$$30 = 7q + r, \quad \text{where } 0 \leq r < 7$$

[10pt]

Let's test values for q: If $q = 1$, $7(1) + r = 30 \Rightarrow 7 + r = 30 \Rightarrow r = 23$. This is not valid as $r < 7$. If $q = 2$, $7(2) + r = 30 \Rightarrow 14 + r = 30 \Rightarrow r = 16$. This is not valid as $r < 7$. If $q = 3$, $7(3) + r = 30 \Rightarrow 21 + r = 30 \Rightarrow r = 9$. This is not valid as $r < 7$. If $q = 4$, $7(4) + r = 30 \Rightarrow 28 + r = 30 \Rightarrow r = 2$. This is valid as $0 \leq 2 < 7$.

[10pt]

So, the quotient is $q = 4$ and the remainder is $r = 2$. This means that 30 days is equal to 4 full weeks and 2 extra days.

[10pt]

To find the day of the week, we add the remainder to the current day's number: Current day (Monday) + Remainder = Day of the week

$$1 + 2 = 3$$

[10pt]

The number 3 corresponds to Wednesday in our representation of the days of the week (Sunday=0, Monday=1, Tuesday=2, Wednesday=3).

[10pt]

Alternatively, we can use modular arithmetic. We want to find the day of the week that is 30 days after Monday. Today is Monday, which we represent as 1. We want to compute $(1 + 30) \pmod 7$.

[10pt]

First, let's find $30 \pmod 7$:

$$30 \equiv r \pmod 7$$

$$30 = 4 \times 7 + 2$$

$$30 \equiv 2 \pmod 7$$

[10pt]

Now substitute this back into the original expression:

$$(1 + 30) \pmod 7 \equiv (1 + 2) \pmod 7$$

$$(1 + 2) \pmod 7 \equiv 3 \pmod 7$$

[10pt]

The result is 3. Since we assigned Sunday = 0, Monday = 1, Tuesday = 2, and Wednesday = 3, the day of the week will be Wednesday.

The final answer is $\boxed{Wednesday}$.

Problem 15

Here's the solution to problem 15:

Problem 15: January 1, 2000, was a Saturday, and 2000 was a leap year. What day of the week will January 1, 2050, be?

Solution:

We need to determine the number of days between January 1, 2000, and January 1, 2050. This period spans exactly 50 years.

First, we need to count the number of leap years between 2000 and 2049, inclusive. A leap year occurs every 4 years, except for years divisible by 100 but not by 400.

The years in the interval are 2000, 2001, ..., 2049.

The leap years in this interval are: * 2000 (since it's divisible by 400) * 2004 * 2008 * 2012 * 2016 * 2020 * 2024 * 2028 * 2032 * 2036 * 2040 * 2044 * 2048

There are 13 leap years between 2000 and 2049, inclusive.

The total number of days in 50 years can be calculated as follows: Number of days = (Number of normal years * 365 days/year) + (Number of leap years * 366 days/year)

Number of normal years = Total years - Number of leap years Number of normal years = 50 - 13 = 37

Total number of days = (37 * 365) + (13 * 366)

Let's calculate this:

$$\text{Total number of days} = (37 \times 365) + (13 \times 366)$$

$$\text{Total number of days} = 13505 + 4758$$

$$\text{Total number of days} = 18263$$

Alternatively, we can calculate the total number of days by considering 50 years of 365 days each, and then adding an extra day for each leap year.

Number of days = (50 years * 365 days/year) + (Number of leap years) Number of days = (50 * 365) + 13

$$\text{Number of days} = 18250 + 13$$

$$\text{Number of days} = 18263$$

Now we need to find out how many weeks are in 18263 days and what the remainder is. We divide the total number of days by 7.

$$18263 \div 7$$

Performing the division: $18263 = 7 \times 2609 + 0$

The remainder is 0. This means that January 1, 2050, will be exactly 2609 weeks after January 1, 2000.

Since January 1, 2000, was a Saturday, and the number of days between January 1, 2000, and January 1, 2050, is an exact multiple of 7, January 1, 2050, will also be a Saturday.

The final answer is $\boxed{Saturday}$.

Problem 16

Problem 16 asks to state the remainder obtained when the quotient-remainder theorem is applied to n with divisor d, given that d is a positive integer and $d|n$.

The quotient-remainder theorem states that for any integer n and any positive integer d, there exist unique integers q (quotient) and r (remainder) such that $n = dq + r$ and $0 \leq r < d$.

We are given that d is a positive integer and $d|n$. By the definition of divisibility, if $d|n$, then there exists an integer k such that $n = dk$.

Now we apply the quotient-remainder theorem to n and d. We need to find integers q and r such that $n = dq + r$ and $0 \leq r < d$.

We have the equation $n = dk$ from the given information. We can write this in the form $n = dq + r$ by setting $q = k$ and $r = 0$.

Let's check if the condition $0 \leq r < d$ is satisfied. Here, $r = 0$. Since d is a positive integer, we know that $d > 0$. Therefore, $0 \leq 0 < d$ is true.

Since $q = k$ is an integer and $r = 0$ satisfies the condition $0 \leq r < d$, the unique integers guaranteed by the quotient-remainder theorem are $q = k$ and $r = 0$.

Thus, the remainder obtained when the quotient-remainder theorem is applied to n with divisor d is 0.

The final answer is $\boxed{0}$.

Problem 17

Here's the solution to problem 17:

17. Prove that the product of any two consecutive integers is even.

Let the two consecutive integers be n and $n+1$, where n is an integer. We want to prove that $n(n+1)$ is even.

We can use the quotient-remainder theorem to consider two cases for the integer n:

Case 1: n is even. If n is even, then by the definition of an even integer, there exists an integer k such that $n = 2k$.

Now, consider the product $n(n+1)$:

$$\begin{aligned} n(n+1) &= (2k)(n+1) \\ &= 2(k(n+1)) \end{aligned}$$

Since k and $n+1$ are integers, their product $k(n+1)$ is also an integer. Let $m = k(n+1)$. Then, $n(n+1) = 2m$, which means that $n(n+1)$ is even.

Case 2: n is odd. If n is odd, then by the definition of an odd integer, there exists an integer k such that $n = 2k+1$.

Now, consider the term $n+1$:

$$\begin{aligned} n+1 &= (2k+1)+1 \\ &= 2k+2 \\ &= 2(k+1) \end{aligned}$$

Since k is an integer, $k+1$ is also an integer. This shows that $n+1$ is even.

Now, consider the product $n(n+1)$:

$$\begin{aligned} n(n+1) &= n(2(k+1)) \\ &= 2(n(k+1)) \end{aligned}$$

Since n and $k+1$ are integers, their product $n(k+1)$ is also an integer. Let $m = n(k+1)$. Then, $n(n+1) = 2m$, which means that $n(n+1)$ is even.

In both cases, we have shown that the product of two consecutive integers is even. Therefore, the product of any two consecutive integers is even.

Problem 18

18. The result of exercise 17 suggests that the second apparent blind alley in the discussion of Example 4.4.7 might not be a blind alley after all. Write a new proof of Theorem 4.4.3 based on this observation.

Theorem 4.4.3: For any integer n, $n^2 - n + 3$ is odd.

Proof:

We will prove this theorem by considering two cases for n: when n is even and when n is odd.

Case 1: n is even. If n is even, then n can be written in the form $n = 2k$ for some integer k.

Substitute this into the expression $n^2 - n + 3$:

We can rewrite this expression as:

Let $m = 2k^2 - k$. Since k is an integer, $2k^2 - k$ is also an integer. Thus, the expression becomes $2m + 3$.

We can further rewrite $2m + 3$ as:

Since m is an integer, $m+1$ is also an integer. Therefore, $2(m+1)+1$ is of the form $2 \times (\text{integer}) + 1$, which means it is odd.

So, if n is even, $n^2 - n + 3$ is odd.

Case 2: n is odd. If n is odd, then n can be written in the form $n = 2k + 1$ for some integer k.

Substitute this into the expression $n^2 - n + 3$:

We can rewrite this expression as:

Let $p = 2k^2 + k$. Since k is an integer, $2k^2 + k$ is also an integer. Thus, the expression becomes $2p+3$.

We can further rewrite $2p + 3$ as:

Since p is an integer, $p + 1$ is also an integer. Therefore, $2(p + 1) + 1$ is of the form $2 \times (\text{integer}) + 1$, which means it is odd.

So, if n is odd, $n^2 - n + 3$ is odd.

Conclusion: In both cases, whether n is even or odd, $n^2 - n + 3$ is odd. Therefore, by the principle of mathematical induction (or in this case, by exhaustion of all possible parities for n), the theorem is proved.

Alternative approach based on Exercise 17:

Exercise 17 states that the product of any two consecutive integers is even. This means that $n(n-1)$ is always even.

Consider the expression $n^2 - n + 3$. We can rewrite this as:

From Exercise 17, we know that $n(n - 1)$ is an even integer. Let $n(n - 1) = 2m$ for some integer m.

Substituting this into the expression:

We can rewrite $2m + 3$ as:

Since m is an integer, $m+1$ is also an integer. Therefore, $2(m+1)+1$ is of the form $2 \times (\text{integer}) + 1$, which means it is odd.

Thus, $n^2 - n + 3$ is always odd.

Problem 19

We are asked to prove that for all integers n, $n^2 - n + 3$ is odd.

Let n be any integer. We will consider two cases based on the parity of n.

Case 1: n is even. If n is even, then n can be written in the form $n = 2k$ for some integer k. Substitute $n = 2k$ into the expression $n^2 - n + 3$:

$$n^2 - n + 3 = (2k)^2 - (2k) + 3$$

$$= 4k^2 - 2k + 3$$

We can rewrite this as:

$$= 2(2k^2 - k) + 3$$

$$= 2(2k^2 - k) + 2 + 1$$

$$= 2(2k^2 - k + 1) + 1$$

Let $m = 2k^2 - k + 1$. Since k is an integer, m is also an integer. Thus, $n^2 - n + 3 = 2m + 1$, which means $n^2 - n + 3$ is odd.

Case 2: n is odd. If n is odd, then n can be written in the form $n = 2k + 1$ for some integer k. Substitute $n = 2k + 1$ into the expression $n^2 - n + 3$:

$$n^2 - n + 3 = (2k + 1)^2 - (2k + 1) + 3$$

$$= (4k^2 + 4k + 1) - 2k - 1 + 3$$

$$= 4k^2 + 4k + 1 - 2k - 1 + 3$$

$$= 4k^2 + 2k + 3$$

We can rewrite this as:

$$= 2(2k^2 + k) + 3$$

$$= 2(2k^2 + k) + 2 + 1$$

$$= 2(2k^2 + k + 1) + 1$$

Let $p = 2k^2 + k + 1$. Since k is an integer, p is also an integer. Thus, $n^2 - n + 3 = 2p + 1$, which means $n^2 - n + 3$ is odd.

In both cases, whether n is even or odd, $n^2 - n + 3$ is odd. Therefore, for all integers n, $n^2 - n + 3$ is odd.

Problem 20

Suppose a is an integer. If $a \pmod 7 = 4$, what is $5a \pmod 7$?

We are given that $a \pmod 7 = 4$. This means that when a is divided by 7, the remainder is 4. According to the definition of the quotient-remainder theorem, this can be written as:

$$a = 7q + 4$$

for some integer q.

We want to find $5a \pmod 7$. First, let's find an expression for $5a$:

$$5a = 5(7q + 4)$$

Now, distribute the 5:

$$5a = 5(7q) + 5(4)$$

$$5a = 35q + 20$$

We want to find the remainder when $5a$ is divided by 7. We can rewrite the expression for $5a$ to isolate a multiple of 7. We can see that $35q$ is already a multiple of 7, since $35 = 7 \times 5$. So, $35q = 7(5q)$.

Now let's consider the term 20. We can divide 20 by 7 to find its remainder:

$$20 = 7 \times 2 + 6$$

So, $20 \pmod 7 = 6$.

Substitute the expression for 20 back into the equation for $5a$:

$$5a = 35q + (7 \times 2 + 6)$$

$$5a = 35q + 14 + 6$$

Now, we can group the terms that are divisible by 7:

$$5a = (35q + 14) + 6$$

$$5a = 7(5q + 2) + 6$$

Let $q' = 5q + 2$. Since q is an integer, $5q$ is an integer, and $5q + 2$ is also an integer. So, we have:

$$5a = 7q' + 6$$

According to the quotient-remainder theorem, this equation means that when $5a$ is divided by 7, the quotient is q' and the remainder is 6. Therefore, $5a \pmod 7 = 6$.

Alternatively, using modular arithmetic properties: We are given $a \equiv 4 \pmod 7$. We want to find $5a \pmod 7$. We can multiply both sides of the congruence by 5:

$$5 \times a \equiv 5 \times 4 \pmod 7$$

$$5a \equiv 20 \pmod 7$$

Now, we find the remainder of 20 when divided by 7:

$$20 = 2 \times 7 + 6$$

So, $20 \equiv 6 \pmod 7$. Substituting this back into the congruence:

$$5a \equiv 6 \pmod 7$$

Thus, $5a \pmod 7 = 6$.

Problem 21

Suppose b is an integer. If $b \pmod{12} = 5$, what is $8b \pmod{12}$?

According to the problem statement, we are given that b is an integer and

$$b \pmod{12} = 5$$

This means that when b is divided by 12, the remainder is 5. By the definition of the quotient-remainder theorem, we can write b in the form:

$$b = 12k + 5$$

for some integer k.

We want to find the value of $8b \pmod{12}$. First, let's find an expression for $8b$ by substituting the expression for b:

$$8b = 8(12k + 5)$$

Now, distribute the 8 to both terms inside the parentheses:

$$8b = 8 \times 12k + 8 \times 5$$

$$8b = 96k + 40$$

We want to find the remainder when $8b$ is divided by 12. We can rewrite the expression for $8b$ in a way that clearly shows a multiple of 12. Notice that $96k$ is a multiple of 12, since $96 = 12 \times 8$. So, we have:

$$8b = 12(8k) + 40$$

Now, we need to find the remainder of 40 when divided by 12. We can write 40 as a multiple of 12 plus a remainder:

$$40 = 12 \times 3 + 4$$

Substitute this expression back into the equation for $8b$:

$$8b = 12(8k) + (12 \times 3 + 4)$$

Now, we can group the terms that are multiples of 12:

$$8b = 12(8k) + 12(3) + 4$$

$$8b = 12(8k + 3) + 4$$

Let $m = 8k + 3$. Since k is an integer, $8k$ is an integer, and $8k + 3$ is also an integer. So, we have:

$$8b = 12m + 4$$

This equation shows that when $8b$ is divided by 12, the quotient is m and the remainder is 4. Therefore, the value of $8b \pmod{12}$ is 4.

$$8b \pmod{12} = 4$$

The final answer is $\boxed{4}$.

Problem 22

Problem 22: Suppose c is an integer. If c mod 12 = 5, what is 10c mod 12? In other words, if division of c by 12 gives a remainder of 5, what is the remainder when 10c is divided by 12?

Given that c is an integer and $c \pmod{12} = 5$. This means that when c is divided by 12, the remainder is 5. According to the quotient-remainder theorem, we can write c as:

$$c = 12q + 5$$

where q is an integer.

We want to find $10c \pmod{12}$. First, let's find an expression for $10c$:

$$10c = 10(12q + 5)$$

Distribute the 10 to both terms inside the parentheses:

$$10c = 10 \times 12q + 10 \times 5$$

$$10c = 120q + 50$$

Now we want to find the remainder when $10c$ is divided by 12. We can rewrite $120q + 50$ in terms of multiples of 12. We can see that $120q$ is already a multiple of 12, since $120 = 12 \times 10$.

$$120q = 12 \times (10q)$$

Now let's consider the term 50. We can divide 50 by 12 to find its remainder:

$$50 = 12 \times 4 + 2$$

So, $50 \equiv 2 \pmod{12}$.

Substitute these back into the expression for $10c$:

$$10c = 12 \times (10q) + (12 \times 4 + 2)$$

$$10c = 12 \times 10q + 12 \times 4 + 2$$

Now we can factor out 12 from the first two terms:

$$10c = 12(10q + 4) + 2$$

Let $q' = 10q + 4$. Since q is an integer, $10q$ is an integer, and $10q + 4$ is also an integer. So, we can write $10c$ in the form $12q' + r$:

$$10c = 12q' + 2$$

According to the quotient-remainder theorem, when $10c$ is divided by 12, the quotient is q' and the remainder is 2. The remainder must satisfy $0 \leq r < 12$. In this case, $r = 2$, which satisfies the condition.

Therefore, $10c \pmod{12} = 2$.

The final answer is $\boxed{2}$.

Problem 23

We want to prove that for all integers n, if $n \pmod 5 = 3$, then $n^2 \pmod 5 = 4$.

Given that n is an integer and $n \pmod 5 = 3$. This means that when n is divided by 5, the remainder is 3. According to the quotient-remainder theorem, we can write n in the form:

$$n = 5q + 3$$

where q is some integer.

Now, we want to find $n^2 \pmod 5$. First, let's find n^2:

$$n^2 = (5q + 3)^2$$

Expand the expression:

$$n^2 = (5q)^2 + 2(5q)(3) + 3^2$$

Perform the multiplication and exponentiation:

$$n^2 = 25q^2 + 30q + 9$$

We want to find the remainder when n^2 is divided by 5. We can rewrite the expression for n^2 by factoring out 5 from the terms that are divisible by 5:

$$n^2 = 5(5q^2) + 5(6q) + 9$$

Now, group the terms that have a factor of 5:

$$n^2 = 5(5q^2 + 6q) + 9$$

Let $k = 5q^2 + 6q$. Since q is an integer, q^2 is an integer, and $5q^2 + 6q$ is also an integer. So, we have:

$$n^2 = 5k + 9$$

We are interested in the remainder when n^2 is divided by 5. The term $5k$ is clearly divisible by 5, so it will have a remainder of 0 when divided by 5. We need to consider the remainder of 9 when divided by 5.

We can write 9 as:

$$9 = 5(1) + 4$$

Substitute this back into the expression for n^2:

$$n^2 = 5k + (5(1) + 4)$$

Rearrange the terms to group the multiples of 5:

$$n^2 = 5k + 5(1) + 4$$

$$n^2 = 5(k + 1) + 4$$

Let $q' = k + 1$. Since k is an integer, q' is also an integer. So, we have:

$$n^2 = 5q' + 4$$

This equation shows that when n^2 is divided by 5, the quotient is q' and the remainder is 4. Therefore, $n^2 \pmod 5 = 4$.

Thus, we have proved that for all integers n, if $n \pmod 5 = 3$, then $n^2 \pmod 5 = 4$.

Problem 24

24. Prove that for all integers m and n, if m mod 5 = 2 and n mod 3 = 6 then mn mod 5 = 1.

The problem statement contains a contradiction. The condition "n mod 3 = 6" is impossible because the remainder when dividing by 3 must be less than 3. Therefore, no integers m and n can satisfy both conditions simultaneously. This means the premise of the implication is false, and thus the implication is vacuously true.

However, assuming there is a typo and "n mod 3 = 6" should be "n mod 3 = some valid remainder", or that the problem intended to be "m mod 5 = 2 and n mod 5 = some value", or "m mod 5 = 2 and n mod X = Y", let's re-examine the problem for potential misinterpretations or common problem types.

Given the context of the surrounding problems that deal with modular arithmetic, it is highly likely there is a typo in "n mod 3 = 6". A common scenario would be that the modulus for n is also 5, or the remainder is a valid one for modulus 3.

Let's assume the question intended to ask: **"Prove that for all integers m and n, if m mod 5 = 2 and n mod 5 = 1, then mn mod 5 = 2."**

Here's the proof for this corrected statement:

By definition of the modulo operation: If $m \pmod 5 = 2$, then there exists an integer k such that

$$m = 5k + 2$$

If $n \pmod 5 = 1$, then there exists an integer l such that

$$n = 5l + 1$$

Now, we want to find the product mn:

$$mn = (5k + 2)(5l + 1)$$

Expand the product:

$$mn = (5k)(5l) + (5k)(1) + (2)(5l) + (2)(1)$$

$$mn = 25kl + 5k + 10l + 2$$

We want to express mn in the form $5q + r$, where $0 \leq r < 5$. We can factor out a 5 from the terms that are multiples of 5:

$$mn = 5(5kl + k + 2l) + 2$$

Let $q = 5kl + k + 2l$. Since k and l are integers, q is also an integer. Therefore, we have:

$$mn = 5q + 2$$

By the definition of the modulo operation, this means:

$$mn \pmod 5 = 2$$

This proves that if $m \pmod 5 = 2$ and $n \pmod 5 = 1$, then $mn \pmod 5 = 2$.

—

Addressing the original problem as stated:

24. Prove that for all integers m and n, if m mod 5 = 2 and n mod 3 = 6 then mn mod 5 = 1.

The statement "n mod 3 = 6" means that when n is divided by 3, the remainder is 6. According to the Quotient-Remainder Theorem, for any integer n and a positive integer d (in this case, $d = 3$), there exist unique integers q (quotient) and r (remainder) such that $n = dq + r$ and $0 \leq r < d$.

For $d = 3$, the possible values for the remainder r are $0, 1, 2$. Since 6 is not in the set $\{0, 1, 2\}$, the condition "n mod 3 = 6" cannot be satisfied by any integer n.

Therefore, the premise "m mod 5 = 2 and n mod 3 = 6" is false.

In logic, an implication of the form "If P, then Q" is considered true if the premise P is false, regardless of the truth value of the conclusion Q. This is known as a vacuous truth.

Since the premise is false, the statement "for all integers m and n, if m mod 5 = 2 and n mod 3 = 6 then mn mod 5 = 1" is vacuously true.

However, it is important to note that this is likely not the intended question due to the impossible condition. If this were a graded problem, it would be advisable to seek clarification from the instructor.

Problem 25

Here's the solution to problem 25:

We want to prove that for all integers a and b, if $a \pmod 7 = 5$ and $b \pmod 7 = 6$, then $ab \pmod 7 = 2$.

Let a and b be integers. We are given that $a \pmod 7 = 5$. This means that when a is divided by 7, the remainder is 5. By the definition of the mod operator, this can be written as:

$$a = 7k + 5$$

for some integer k.

We are also given that $b \pmod 7 = 6$. This means that when b is divided by 7, the remainder is 6. By the definition of the mod operator, this can be written as:

$$b = 7m + 6$$

for some integer m.

Now we want to find the product ab.

$$ab = (7k + 5)(7m + 6)$$

Expand the product:

$$ab = 7k(7m + 6) + 5(7m + 6)$$

$$ab = 49km + 42k + 35m + 30$$

We want to find the remainder when ab is divided by 7. We can group the terms that are divisible by 7:

$$ab = (49km + 42k + 35m) + 30$$

$$ab = 7(7km + 6k + 5m) + 30$$

Let $p = 7km + 6k + 5m$. Since k and m are integers, p is also an integer. So, we have:

$$ab = 7p + 30$$

Now we need to find the remainder of 30 when divided by 7.

$$30 = 4 \times 7 + 2$$

Substitute this back into the expression for ab:

$$ab = 7p + (4 \times 7 + 2)$$

$$ab = 7p + 7 \times 4 + 2$$

Factor out 7 from the first two terms:

$$ab = 7(p + 4) + 2$$

Let $q = p + 4$. Since p is an integer, q is also an integer. Thus, we can write:

$$ab = 7q + 2$$

This equation shows that when ab is divided by 7, the quotient is q and the remainder is 2. Therefore, by the definition of the mod operator:

$$ab \pmod 7 = 2$$

This proves that if $a \pmod 7 = 5$ and $b \pmod 7 = 6$, then $ab \pmod 7 = 2$.

Problem 26

26. Prove that a necessary and sufficient condition for a non-negative integer n to be divisible by a positive integer d is that $n \pmod{d} = 0$.

We need to prove two implications: 1. If n is divisible by d, then $n \pmod{d} = 0$. (Necessity) 2. If $n \pmod{d} = 0$, then n is divisible by d. (Sufficiency)

Part 1: Necessity

Assume n is a non-negative integer and d is a positive integer such that n is divisible by d.

By the definition of divisibility, if n is divisible by d, then there exists an integer q such that $n = dq$.

$$n = dq$$

According to the quotient-remainder theorem, for any non-negative integer n and positive integer d, there exist unique integers q' and r such that $n = dq' + r$ and $0 \leq r < d$.

Comparing this general form with our specific equation $n = dq$:

$$\begin{aligned} dq &= dq' + r \\ 0 &= dq' - dq + r \\ 0 &= d(q' - q) + r \end{aligned}$$

Let $q'' = q' - q$. Since q' and q are integers, q'' is also an integer.

$$\begin{aligned} 0 &= dq'' + r \\ r &= -dq'' \end{aligned}$$

Since $d > 0$ and q'' is an integer, dq'' is a multiple of d. From $r = -dq''$, we know that r is also a multiple of d.

We also have the condition from the quotient-remainder theorem that $0 \leq r < d$. The only multiple of d that satisfies $0 \leq r < d$ is 0. Therefore, $r = 0$.

Since $n \pmod{d}$ is defined as the remainder r when n is divided by d, we have:

$$n \pmod{d} = r = 0$$

Thus, if n is divisible by d, then $n \pmod{d} = 0$.

Part 2: Sufficiency

Assume n is a non-negative integer and d is a positive integer such that $n \pmod{d} = 0$.

By the definition of the modulo operation, $n \pmod{d}$ is the remainder when n is divided by d. So, $n \pmod{d} = 0$ means that the remainder is 0.

According to the quotient-remainder theorem, there exist unique integers q and r such that $n = dq + r$ and $0 \leq r < d$.

Given that $n \pmod{d} = 0$, we know that the remainder r is 0.

$$r = 0$$

Substituting $r = 0$ into the equation from the quotient-remainder theorem:

$$n = dq + 0$$
$$n = dq$$

By the definition of divisibility, if there exists an integer q such that $n = dq$, then n is divisible by d. Since we found such an integer q (which is the quotient from the quotient-remainder theorem), n is divisible by d.

Thus, if $n \pmod{d} = 0$, then n is divisible by d.

Conclusion:

We have shown both that if n is divisible by d, then $n \pmod{d} = 0$, and that if $n \pmod{d} = 0$, then n is divisible by d. Therefore, a necessary and sufficient condition for a non-negative integer n to be divisible by a positive integer d is that $n \pmod{d} = 0$.

Problem 27

Show that any integer n can be written in one of the three forms $n = 3q$ or $n = 3q + 1$ or $n = 3q + 2$ for some integer q.

By the Quotient-Remainder Theorem, for any integer n and any positive integer d, there exist unique integers q and r such that $n = dq + r$ and $0 \leq r < d$. [10pt] In this problem, we are considering the integer n and the divisor $d = 3$. [10pt] According to the Quotient-Remainder Theorem, there exist unique integers q and r such that $n = 3q + r$, where $0 \leq r < 3$. [10pt] The possible values for the remainder r are 0, 1, and 2. [10pt] We consider each possible value of r separately: [10pt] Case 1: $r = 0$. [10pt] In this case, $n = 3q + 0$, which simplifies to $n = 3q$. [10pt] Case 2: $r = 1$. [10pt] In this case, $n = 3q + 1$. [10pt] Case 3: $r = 2$. [10pt] In this case, $n = 3q + 2$. [10pt] Since these are all the possible values for the remainder r, any integer n must be of the form $3q$, $3q + 1$, or $3q + 2$ for some integer q. [10pt] Therefore, any integer n can be written in one of the three forms $n = 3q$ or $n = 3q + 1$ or $n = 3q + 2$ for some integer q.

Problem 28

Here's the solution to problem 28:

28. a. Use the quotient-remainder theorem with d = 3 to prove that the product of any three consecutive integers is divisible by 3.

Let the three consecutive integers be n, $n+1$, and $n+2$. By the quotient-remainder theorem, when n is divided by 3, there exist unique integers q and r such that $n = 3q + r$, where $0 \leq r < 3$. This means r can be 0, 1, or 2.

We will consider these three cases for r:

Case 1: $r = 0$ If $r = 0$, then $n = 3q$. In this case, the product of the three consecutive integers is:

$$n(n+1)(n+2) = (3q)(3q+1)(3q+2)$$

$$n(n+1)(n+2) = 3[q(3q+1)(3q+2)]$$

Since q, $3q+1$, and $3q+2$ are integers, their product $q(3q+1)(3q+2)$ is also an integer. Let $k = q(3q+1)(3q+2)$. Then, $n(n+1)(n+2) = 3k$. This shows that the product is divisible by 3.

Case 2: $r = 1$ If $r = 1$, then $n = 3q+1$. In this case, the three consecutive integers are $3q+1$, $(3q+1)+1 = 3q+2$, and $(3q+1)+2 = 3q+3$. The product is:

$$n(n+1)(n+2) = (3q+1)((3q+1)+1)((3q+1)+2)$$

$$n(n+1)(n+2) = (3q+1)(3q+2)(3q+3)$$

We can factor out a 3 from the third term:

$$n(n+1)(n+2) = (3q+1)(3q+2) \cdot 3(q+1)$$

$$n(n+1)(n+2) = 3[(3q+1)(3q+2)(q+1)]$$

Since $3q+1$, $3q+2$, and $q+1$ are integers, their product $(3q+1)(3q+2)(q+1)$ is also an integer. Let $k = (3q+1)(3q+2)(q+1)$. Then, $n(n+1)(n+2) = 3k$. This shows that the product is divisible by 3.

Case 3: $r = 2$ If $r = 2$, then $n = 3q+2$. In this case, the three consecutive integers are $3q+2$, $(3q+2)+1 = 3q+3$, and $(3q+2)+2 = 3q+4$. The product is:

$$n(n+1)(n+2) = (3q+2)((3q+2)+1)((3q+2)+2)$$

$$n(n+1)(n+2) = (3q+2)(3q+3)(3q+4)$$

We can factor out a 3 from the second term:

$$n(n+1)(n+2) = (3q+2) \cdot 3(q+1) \cdot (3q+4)$$

$$n(n+1)(n+2) = 3[(3q+2)(q+1)(3q+4)]$$

Since $3q+2$, $q+1$, and $3q+4$ are integers, their product $(3q+2)(q+1)(3q+4)$ is also an integer. Let $k = (3q+2)(q+1)(3q+4)$. Then, $n(n+1)(n+2) = 3k$. This shows that the product is divisible by 3.

In all three possible cases for the remainder when n is divided by 3, the product of the three consecutive integers n, $n+1$, and $n+2$ is divisible by 3.

b. Use the mod notation to rewrite the result of part (a).

The result of part (a) states that the product of any three consecutive integers is divisible by 3. In

modular arithmetic notation, a number a is divisible by b if and only if $a \pmod{b} = 0$. Therefore, we can rewrite the result as:

$$n(n+1)(n+2) \equiv 0 \pmod{3}$$

for any integer n.

Problem 29

Here's the solution to problem 29:

29. a. Use the quotient-remainder theorem with d = 3 to prove that the square of any integer has the form 3k or 3k + 1 for some integer k.

We use the quotient-remainder theorem with the divisor $d = 3$. For any integer n, there exist unique integers q and r such that $n = 3q + r$ and $0 \leq r < 3$.

The possible values for the remainder r are 0, 1, and 2. We consider each case for the square of n, which is n^2.

Case 1: $r = 0$ If $r = 0$, then $n = 3q$. Squaring both sides, we get:

$$n^2 \overset{*}{=} (3q)^2$$

$$n^2 \overset{*}{=} 9q^2$$

We can rewrite $9q^2$ as $3(3q^2)$. Let $k = 3q^2$. Since q is an integer, q^2 is an integer, and $3q^2$ is also an integer. Therefore, $n^2 = 3k$ for some integer k.

Case 2: $r = 1$ If $r = 1$, then $n = 3q + 1$. Squaring both sides, we get:

$$n^2 \overset{*}{=} (3q + 1)^2$$

$$n^2 \overset{*}{=} (3q)^2 + 2(3q)(1) + 1^2$$

$$n^2 \overset{*}{=} 9q^2 + 6q + 1$$

We can group the terms with q and factor out a 3:

$$n^2 \overset{*}{=} 3(3q^2 + 2q) + 1$$

Let $k = 3q^2 + 2q$. Since q is an integer, q^2 is an integer, $3q^2$ is an integer, $2q$ is an integer, and their sum $3q^2 + 2q$ is also an integer. Therefore, $n^2 = 3k + 1$ for some integer k.

Case 3: $r = 2$ If $r = 2$, then $n = 3q + 2$. Squaring both sides, we get:

$$n^2 \overset{*}{=} (3q + 2)^2$$

$$n^2 \overset{*}{=} (3q)^2 + 2(3q)(2) + 2^2$$

$$n^2 \overset{*}{=} 9q^2 + 12q + 4$$

We can rewrite 4 as $3 + 1$ and then group terms to factor out a 3:

$$n^2 \overset{*}{=} 9q^2 + 12q + 3 + 1$$

$$n^2 \overset{*}{=} 3(3q^2 + 4q + 1) + 1$$

Let $k = 3q^2 + 4q + 1$. Since q is an integer, q^2 is an integer, $3q^2$ is an integer, $4q$ is an integer, and their sum $3q^2 + 4q + 1$ is also an integer. Therefore, $n^2 = 3k + 1$ for some integer k.

In all possible cases ($r = 0$, $r = 1$, and $r = 2$), the square of an integer n can be written in the form $3k$ or $3k + 1$ for some integer k.

b. Use the mod notation to rewrite the result of part (a).

From part (a), we showed that for any integer n, n^2 is either of the form $3k$ or $3k + 1$.

In terms of modular arithmetic: - If $n^2 = 3k$, then $n^2 \equiv 0 \pmod 3$. - If $n^2 = 3k + 1$, then $n^2 \equiv 1 \pmod 3$.

Therefore, the result of part (a) can be rewritten using mod notation as: For any integer n, $n^2 \equiv 0 \pmod 3$ or $n^2 \equiv 1 \pmod 3$.

Problem 30

The problem asks to prove that the product of any two consecutive integers has the form $3k$ or $3k+2$ for some integer k.

We can use the quotient-remainder theorem to analyze this problem. The quotient-remainder theorem states that for any integer n and any positive integer d, there exist unique integers q and r such that $n = dq + r$ and $0 \leq r < d$.

In this case, we are considering any integer n. We can divide n by 3. According to the quotient-remainder theorem, when n is divided by 3, there are three possible remainders: 0, 1, or 2. So, n can be written in one of the following three forms:

$$\begin{aligned} n &= 3q \\ n &= 3q+1 \\ n &= 3q+2 \end{aligned}$$

for some integer q.

Now, let's consider the product of two consecutive integers. Let these consecutive integers be n and $n+1$. We need to examine the form of $n(n+1)$ based on the possible forms of n.

Case 1: $n = 3q$. In this case, the product of the two consecutive integers is:

$$\begin{aligned} n(n+1) &= (3q)(3q+1) \\ &= 9q^2 + 3q \\ &= 3(3q^2 + q) \end{aligned}$$

Let $k = 3q^2 + q$. Since q is an integer, q^2 is an integer, $3q^2$ is an integer, and $3q^2 + q$ is an integer. Thus, $n(n+1) = 3k$. This is of the form $3k$.

Case 2: $n = 3q+1$. In this case, the product of the two consecutive integers is:

$$\begin{aligned} n(n+1) &= (3q+1)((3q+1)+1) \\ &= (3q+1)(3q+2) \\ &= (3q)(3q) + (3q)(2) + (1)(3q) + (1)(2) \\ &= 9q^2 + 6q + 3q + 2 \\ &= 9q^2 + 9q + 2 \\ &= 3(3q^2 + 3q) + 2 \end{aligned}$$

Let $k = 3q^2 + 3q$. Since q is an integer, q^2 is an integer, $3q^2$ is an integer, $3q$ is an integer, and $3q^2 + 3q$ is an integer. Thus, $n(n+1) = 3k+2$. This is of the form $3k+2$.

Case 3: $n = 3q+2$. In this case, the product of the two consecutive integers is:

$$\begin{aligned} n(n+1) &= (3q+2)((3q+2)+1) \\ &= (3q+2)(3q+3) \\ &= (3q+2) \cdot 3(q+1) \\ &= 3 \cdot (3q+2)(q+1) \end{aligned}$$

Let $k = (3q+2)(q+1)$. Since q is an integer, $q+1$ is an integer, $3q+2$ is an integer, and $(3q+2)(q+1)$ is an integer. Thus, $n(n+1) = 3k$. This is of the form $3k$.

We have shown that for any integer n, the product $n(n+1)$ is either of the form $3k$ or $3k+2$ for some integer k.

Part b of problem 30 asks to rewrite the result using mod notation. We have shown that $n(n+1)$ can be $3k$ or $3k+2$. This means that $n(n+1) \equiv 0 \pmod 3$ or $n(n+1) \equiv 2 \pmod 3$.

The possible remainders when the product of two consecutive integers is divided by 3 are 0 and 2.

The final answer is $\boxed{3k \text{ or } 3k+2}$.

Problem 31

Here's the solution to problem 31:

31. **a.** Prove that for all integers m and n, $m+n$ and $m-n$ are either both odd or both even.

We will consider four cases based on the parity of m and n.

Case 1: m is even and n is even. If m is even, then $m = 2k$ for some integer k. If n is even, then $n = 2l$ for some integer l.

Then, $m+n = 2k+2l = 2(k+l)$. Since $k+l$ is an integer, $m+n$ is even. Also, $m-n = 2k-2l = 2(k-l)$. Since $k-l$ is an integer, $m-n$ is even. In this case, both $m+n$ and $m-n$ are even.

Case 2: m is odd and n is odd. If m is odd, then $m = 2k+1$ for some integer k. If n is odd, then $n = 2l+1$ for some integer l.

Then, $m+n = (2k+1)+(2l+1) = 2k+2l+2 = 2(k+l+1)$. Since $k+l+1$ is an integer, $m+n$ is even. Also, $m-n = (2k+1)-(2l+1) = 2k+1-2l-1 = 2k-2l = 2(k-l)$. Since $k-l$ is an integer, $m-n$ is even. In this case, both $m+n$ and $m-n$ are even.

Case 3: m is even and n is odd. If m is even, then $m = 2k$ for some integer k. If n is odd, then $n = 2l+1$ for some integer l.

Then, $m+n = 2k+(2l+1) = 2k+2l+1 = 2(k+l)+1$. Since $k+l$ is an integer, $m+n$ is odd. Also, $m-n = 2k-(2l+1) = 2k-2l-1 = 2(k-l)-1 = 2(k-l-1)+2-1 = 2(k-l-1)+1$. Since $k-l-1$ is an integer, $m-n$ is odd. In this case, both $m+n$ and $m-n$ are odd.

Case 4: m is odd and n is even. If m is odd, then $m = 2k+1$ for some integer k. If n is even, then $n = 2l$ for some integer l.

Then, $m+n = (2k+1)+2l = 2k+2l+1 = 2(k+l)+1$. Since $k+l$ is an integer, $m+n$ is odd. Also, $m-n = (2k+1)-2l = 2k-2l+1 = 2(k-l)+1$. Since $k-l$ is an integer, $m-n$ is odd. In this case, both $m+n$ and $m-n$ are odd.

In all four cases, $m+n$ and $m-n$ are either both even or both odd. Therefore, for all integers m and n, $m+n$ and $m-n$ are either both odd or both even.

b. Find all solutions to the equation $m^2 - n^2 = 56$ for which both m and n are positive integers.

We can factor the left side of the equation:

$$m^2 - n^2 = (m-n)(m+n)$$

So, we have:

$$(m-n)(m+n) = 56$$

Since m and n are positive integers, $m+n$ is a positive integer. Also, since $(m-n)(m+n) = 56$ (which is positive), $m-n$ must also be a positive integer. This implies $m > n$.

Let $a = m-n$ and $b = m+n$. We have $ab = 56$, and since m and n are positive integers, $b > a > 0$. Also, from part (a), we know that $m+n$ and $m-n$ have the same parity. Since their product 56 is even, both $m-n$ and $m+n$ must be even.

We need to find pairs of even factors of 56 such that the first factor is smaller than the second. The factors of 56 are 1, 2, 4, 7, 8, 14, 28, 56.

The pairs of even factors are: 1. $m-n = 2$, $m+n = 28$ 2. $m-n = 4$, $m+n = 14$ 3. $m-n = 8$, $m+n = 7$ (This case is not possible because $m-n$ must be less than $m+n$)

Let's solve the systems of equations for the valid pairs:

System 1: $m-n = 2$ $m+n = 28$ Adding the two equations: $(m-n)+(m+n) = 2+28$ $2m = 30$ $m = 15$ Substitute $m = 15$ into the second equation: $15+n = 28$ $n = 28-15$ $n = 13$ Check if m and n are positive: $m = 15 > 0$, $n = 13 > 0$. This is a valid solution.

System 2: $m - n = 4$ $m + n = 14$ Adding the two equations: $(m - n) + (m + n) = 4 + 14$ $2m = 18$ $m = 9$ Substitute $m = 9$ into the second equation: $9 + n = 14$ $n = 14 - 9$ $n = 5$ Check if m and n are positive: $m = 9 > 0$, $n = 5 > 0$. This is a valid solution.

The solutions (m, n) for which both m and n are positive integers are $(15, 13)$ and $(9, 5)$.

c. Find all solutions to the equation $m^2 - n^2 = 88$ for which both m and n are positive integers.

Similar to part (b), we factor the equation:

$$(m - n)(m + n) = 88$$

Since m and n are positive integers, $m + n$ is a positive integer. Since their product is positive, $m - n$ must also be a positive integer, implying $m > n$. Let $a = m - n$ and $b = m + n$. We have $ab = 88$, and $b > a > 0$. From part (a), $m - n$ and $m + n$ must have the same parity. Since their product 88 is even, both $m - n$ and $m + n$ must be even.

We need to find pairs of even factors of 88 such that the first factor is smaller than the second. The factors of 88 are 1, 2, 4, 8, 11, 22, 44, 88.

The pairs of even factors are: 1. $m - n = 2$, $m + n = 44$ 2. $m - n = 4$, $m + n = 22$ 3. $m - n = 8$, $m + n = 11$ (This case is not possible because $m - n$ must be even)

Let's solve the systems of equations for the valid pairs:

System 1: $m - n = 2$ $m + n = 44$ Adding the two equations: $(m - n) + (m + n) = 2 + 44$ $2m = 46$ $m = 23$ Substitute $m = 23$ into the second equation: $23 + n = 44$ $n = 44 - 23$ $n = 21$ Check if m and n are positive: $m = 23 > 0$, $n = 21 > 0$. This is a valid solution.

System 2: $m - n = 4$ $m + n = 22$ Adding the two equations: $(m - n) + (m + n) = 4 + 22$ $2m = 26$ $m = 13$ Substitute $m = 13$ into the second equation: $13 + n = 22$ $n = 22 - 13$ $n = 9$ Check if m and n are positive: $m = 13 > 0$, $n = 9 > 0$. This is a valid solution.

The solutions (m, n) for which both m and n are positive integers are $(23, 21)$ and $(13, 9)$.

Problem 32

Here's the solution to problem 32:

32. Given any integers a, b, and c, if ab is even and $b - c$ is even, what can you say about the parity of $2a - (b + c)$? Prove your answer.

Proof:

We are given that ab is even. This means that either a is even or b is even (or both).

We are also given that $b - c$ is even. This implies that b and c have the same parity, meaning they are both even or both odd.

We want to determine the parity of $2a - (b + c)$.

First, consider the term $2a$. Since a is an integer, $2a$ is always an even number, regardless of the parity of a.

Now consider the term $b + c$. Case 1: b is even. Since $b - c$ is even, and b is even, c must also be even. In this case, $b + c$ is the sum of two even numbers, which is even.

Case 2: b is odd. Since $b - c$ is even, and b is odd, c must also be odd. In this case, $b + c$ is the sum of two odd numbers, which is even.

In both cases, $b + c$ is even.

Now we have $2a - (b + c)$. We know that $2a$ is even. We have shown that $b + c$ is even. Therefore, $2a - (b + c)$ is the difference of two even numbers. The difference of two even numbers is always even.

Thus, $2a - (b + c)$ is even.

Problem 33

Here's the solution to problem 33:

Problem 33: Given any integers a, b, and c, if ab is odd and $b - c$ is even, what can you say about the parity of $a \cdot c$? Prove your answer.

Solution:

We are given that ab is odd. [10pt] For the product of two integers to be odd, both integers must be odd. [10pt] Therefore, a is odd and b is odd. [10pt] We are also given that $b - c$ is even. [10pt] For the difference of two integers to be even, both integers must have the same parity (both even or both odd). [10pt] Since we have established that b is odd, c must also be odd for $b - c$ to be even. [10pt] Now we need to determine the parity of $a \cdot c$. [10pt] We know that a is odd and c is odd. [10pt] The product of two odd integers is always odd. [10pt] Thus, $a \cdot c$ is odd.

Proof: [10pt] Let a, b, c be integers. [10pt] Assume ab is odd. [10pt] By the definition of an odd product, this implies that a is odd and b is odd. [10pt] Assume $b - c$ is even. [10pt] By the definition of an even difference, this implies that b and c have the same parity. [10pt] Since we know b is odd, it follows that c must also be odd. [10pt] We want to determine the parity of $a \cdot c$. [10pt] We have established that a is odd and c is odd. [10pt] The product of two odd integers is odd. [10pt] Therefore, $a \cdot c$ is odd.

Problem 34

Here's the solution to problem 34:

34. Given any integer n, if n ¿ 3, could n, n + 2, and n + 4 all be prime? Prove or give a counterexample.

We need to determine if it's possible for three numbers of the form n, n+2, and n+4 to all be prime, given that n ¿ 3.

Consider the possible remainders when an integer is divided by 3. Any integer n can be written in one of the following forms: $n = 3q$, $n = 3q + 1$, or $n = 3q + 2$ for some integer q.

We will examine each of these cases for the integer n.

Case 1: n is divisible by 3. If $n = 3q$, and since we are given that $n > 3$, then n must be a multiple of 3 greater than 3. The only prime number that is a multiple of 3 is 3 itself. Therefore, if n is divisible by 3 and $n > 3$, n cannot be prime. This means that n, n+2, and n+4 cannot all be prime in this case.

Case 2: n has a remainder of 1 when divided by 3. If $n = 3q + 1$.

Consider the number $n + 2$. $n + 2 = (3q + 1) + 2$ $n + 2 = 3q + 3$ $n + 2 = 3(q + 1)$

This shows that $n + 2$ is divisible by 3. Since we are considering the case where n ¿ 3, it implies that $n + 2 > 5$. If $n + 2$ is divisible by 3 and $n + 2 > 3$, then $n + 2$ cannot be prime. Thus, in this case, n+2 is not prime.

Case 3: n has a remainder of 2 when divided by 3. If $n = 3q + 2$.

Consider the number $n + 4$. $n + 4 = (3q + 2) + 4$ $n + 4 = 3q + 6$ $n + 4 = 3(q + 2)$

This shows that $n + 4$ is divisible by 3. Since we are considering the case where n ¿ 3, it implies that $n + 4 > 7$. If $n + 4$ is divisible by 3 and $n + 4 > 3$, then $n + 4$ cannot be prime. Thus, in this case, n+4 is not prime.

In all possible cases for an integer n divided by 3, at least one of the numbers n, n+2, or n+4 is divisible by 3. For these numbers to all be prime, they would all have to be equal to 3, which is impossible since they are distinct values ($n < n + 2 < n + 4$).

Let's consider the specific example where $n = 3$. In this case, the numbers are 3, 3+2=5, and 3+4=7. All three are prime. However, the problem states "if n ¿ 3".

Therefore, if n ¿ 3, it is not possible for n, n+2, and n+4 to all be prime.

Counterexample: Let $n = 4$. n is not prime. The set is 4, 6, 8. None are prime. Let $n = 5$. n is prime. The set is 5, 7, 9. 9 is not prime. Let $n = 7$. n is prime. The set is 7, 9, 11. 9 is not prime. Let $n = 9$. n is not prime. The set is 9, 11, 13. 9 is not prime.

The proof above demonstrates that for any integer $n > 3$, one of n, $n + 2$, or $n + 4$ must be divisible by 3 and greater than 3, and therefore not prime.

Conclusion: No, it is not possible for n, n+2, and n+4 to all be prime if n ¿ 3.

Problem 35

Here's the solution to problem 35:

Problem 35: The fourth power of any integer has the form $8m$ or $8m+1$ for some integer m.

Proof:

We will use the quotient-remainder theorem to consider all possible remainders when an integer is divided by 8. Let n be any integer. By the quotient-remainder theorem, there exist unique integers q and r such that $n = 8q + r$, where $0 \leq r < 8$. This means r can be $0, 1, 2, 3, 4, 5, 6$, or 7.

We will consider the fourth power of n, which is n^4, for each possible value of r.

Case 1: $r = 0$ If $r = 0$, then $n = 8q$. $n^4 = (8q)^4 = 8^4q^4 = 4096q^4$. We can write $n^4 = 8(512q^4)$. Let $m = 512q^4$. Since q is an integer, q^4 is an integer, and therefore m is an integer. So, $n^4 = 8m$.

[10pt]

Case 2: $r = 1$ If $r = 1$, then $n = 8q + 1$. $n^4 = (8q + 1)^4$. We can use the binomial expansion, but it's easier to see a pattern by considering $(x + 1)^4$. Let's consider $(8q + 1)^4$. $n^4 = ((8q + 1)^2)^2 = (64q^2 + 16q + 1)^2$. We can group the terms as $(64q^2 + 16q) + 1$. $n^4 = ((64q^2 + 16q) + 1)^2 = (64q^2 + 16q)^2 + 2(64q^2 + 16q)(1) + 1^2$. $n^4 = (64q^2 + 16q)^2 + 128q^2 + 32q + 1$. Let's factor out 8 from the first two terms of the expansion: $n^4 = (8(8q^2 + 2q))^2 + 128q^2 + 32q + 1$. $n^4 = 64(8q^2 + 2q)^2 + 128q^2 + 32q + 1$. $n^4 = 8[8(8q^2 + 2q)^2 + 16q^2 + 4q] + 1$. Let $m = 8(8q^2 + 2q)^2 + 16q^2 + 4q$. Since q is an integer, m is an integer. So, $n^4 = 8m + 1$.

Alternatively, and more simply, we can consider the remainders modulo 8. If $n \equiv 1 \pmod 8$, then $n^4 \equiv 1^4 \pmod 8$, which means $n^4 \equiv 1 \pmod 8$. This implies n^4 can be written in the form $8m + 1$.

[10pt]

Case 3: $r = 2$ If $r = 2$, then $n = 8q + 2$. $n^4 = (8q + 2)^4 = (2(4q + 1))^4 = 16(4q + 1)^4$. $n^4 = 16 \times (\text{some integer})$. Since 16 is a multiple of 8 ($16 = 8 \times 2$), n^4 is a multiple of 8. $n^4 = 8(2(4q+1)^4)$. Let $m = 2(4q + 1)^4$. Since q is an integer, m is an integer. So, $n^4 = 8m$.

Alternatively, $n \equiv 2 \pmod 8$. $n^4 \equiv 2^4 \pmod 8 \equiv 16 \pmod 8 \equiv 0 \pmod 8$. This implies n^4 can be written in the form $8m$.

[10pt]

Case 4: $r = 3$ If $r = 3$, then $n = 8q + 3$. $n \equiv 3 \pmod 8$. $n^4 \equiv 3^4 \pmod 8 \equiv 81 \pmod 8$. $81 = 10 \times 8 + 1$. So, $n^4 \equiv 1 \pmod 8$. This implies n^4 can be written in the form $8m + 1$.

[10pt]

Case 5: $r = 4$ If $r = 4$, then $n = 8q + 4$. $n^4 = (8q + 4)^4 = (4(2q + 1))^4 = 256(2q + 1)^4$. Since 256 is a multiple of 8 ($256 = 8 \times 32$), n^4 is a multiple of 8. $n^4 = 8(32(2q + 1)^4)$. Let $m = 32(2q + 1)^4$. Since q is an integer, m is an integer. So, $n^4 = 8m$.

Alternatively, $n \equiv 4 \pmod 8$. $n^4 \equiv 4^4 \pmod 8 \equiv 256 \pmod 8 \equiv 0 \pmod 8$. This implies n^4 can be written in the form $8m$.

[10pt]

Case 6: $r = 5$ If $r = 5$, then $n = 8q + 5$. $n \equiv 5 \pmod 8$. $n^4 \equiv 5^4 \pmod 8 \equiv 625 \pmod 8$. $625 = 78 \times 8 + 1$. So, $n^4 \equiv 1 \pmod 8$. This implies n^4 can be written in the form $8m + 1$.

[10pt]

Case 7: $r = 6$ If $r = 6$, then $n = 8q + 6$. $n^4 = (8q + 6)^4 = (2(4q + 3))^4 = 16(4q + 3)^4$. Since 16 is a multiple of 8 ($16 = 8 \times 2$), n^4 is a multiple of 8. $n^4 = 8(2(4q + 3)^4)$. Let $m = 2(4q + 3)^4$. Since q is an integer, m is an integer. So, $n^4 = 8m$.

Alternatively, $n \equiv 6 \pmod 8$. $n^4 \equiv 6^4 \pmod 8 \equiv 1296 \pmod 8$. $1296 = 162 \times 8 + 0$. So, $n^4 \equiv 0 \pmod 8$. This implies n^4 can be written in the form $8m$.

[10pt]

Case 8: $r = 7$ If $r = 7$, then $n = 8q + 7$. $n \equiv 7 \pmod 8$, which is equivalent to $n \equiv -1 \pmod 8$. $n^4 \equiv (-1)^4 \pmod 8 \equiv 1 \pmod 8$. This implies n^4 can be written in the form $8m + 1$.

[10pt]

In all possible cases for the remainder r, we have shown that n^4 is either of the form $8m$ or $8m+1$ for some integer m. Therefore, the fourth power of any integer has the form $8m$ or $8m + 1$ for some integer m.

■

Problem 36

Here's the solution to problem 36, showing every algebraic step clearly:

Problem 36: The product of any four consecutive integers is divisible by 8.

Proof: Let the four consecutive integers be n, $n+1$, $n+2$, and $n+3$. Their product is $P = n(n+1)(n+2)(n+3)$.

We need to show that P is divisible by 8. We can consider the possible forms of the integer n based on its remainder when divided by 4.

Case 1: n is a multiple of 4. If n is a multiple of 4, then $n = 4k$ for some integer k.

$$P = (4k)(4k+1)(4k+2)(4k+3)$$

We can factor out a 2 from the third term:

$$P = (4k)(4k+1)(2(2k+1))(4k+3)$$

$$P = 8k(4k+1)(2k+1)(4k+3)$$

Since k, $(4k+1)$, $(2k+1)$, and $(4k+3)$ are integers, P is a multiple of 8.

Case 2: n has a remainder of 1 when divided by 4. If n has a remainder of 1 when divided by 4, then $n = 4k+1$ for some integer k.

$$P = (4k+1)(4k+1+1)(4k+1+2)(4k+1+3)$$

$$P = (4k+1)(4k+2)(4k+3)(4k+4)$$

We can factor out a 2 from the second term and a 4 from the fourth term:

$$P = (4k+1)(2(2k+1))(4k+3)(4(k+1))$$

$$P = 2 \cdot 4 \cdot (4k+1)(2k+1)(4k+3)(k+1)$$

$$P = 8(4k+1)(2k+1)(4k+3)(k+1)$$

Since $(4k+1)$, $(2k+1)$, $(4k+3)$, and $(k+1)$ are integers, P is a multiple of 8.

Case 3: n has a remainder of 2 when divided by 4. If n has a remainder of 2 when divided by 4, then $n = 4k+2$ for some integer k.

$$P = (4k+2)(4k+2+1)(4k+2+2)(4k+2+3)$$

$$P = (4k+2)(4k+3)(4k+4)(4k+5)$$

We can factor out a 2 from the first term and a 4 from the third term:

$$P = (2(2k+1))(4k+3)(4(k+1))(4k+5)$$

$$P = 2 \cdot 4 \cdot (2k+1)(4k+3)(k+1)(4k+5)$$

$$P = 8(2k+1)(4k+3)(k+1)(4k+5)$$

Since $(2k+1)$, $(4k+3)$, $(k+1)$, and $(4k+5)$ are integers, P is a multiple of 8.

Case 4: n has a remainder of 3 when divided by 4. If n has a remainder of 3 when divided by 4,

then $n = 4k + 3$ for some integer k.

$$P = (4k + 3)(4k + 3 + 1)(4k + 3 + 2)(4k + 3 + 3)$$

$$P = (4k + 3)(4k + 4)(4k + 5)(4k + 6)$$

We can factor out a 4 from the second term and a 2 from the fourth term:

$$P = (4k + 3)(4(k + 1))(4k + 5)(2(2k + 3))$$

$$P = 4 \cdot 2 \cdot (4k + 3)(k + 1)(4k + 5)(2k + 3)$$

$$P = 8(4k + 3)(k + 1)(4k + 5)(2k + 3)$$

Since $(4k + 3)$, $(k + 1)$, $(4k + 5)$, and $(2k + 3)$ are integers, P is a multiple of 8.

In all possible cases, the product of any four consecutive integers is divisible by 8.

Alternatively, we can reason as follows: In any set of four consecutive integers, there must be at least one multiple of 4. There must also be at least two even numbers. One of these even numbers is a multiple of 4. The other even number is of the form $4m + 2$, which is $2(2m + 1)$. Thus, the product of the two even numbers is of the form $(4m) \cdot 2(2m + 1) = 8m(2m + 1)$ or $2(2m + 1) \cdot 4m' = 8m'(2m + 1)$, which is always divisible by 8. Since the product of all four integers includes the product of these two even numbers, the entire product must be divisible by 8.

The final answer is $\boxed{Theproductofanyfourconsecutiveintegersisdivisibleby8}$.

Problem 37

Prove that the square of any integer has the form $4k$ or $4k+1$ for some integer k.

Let n be any integer. According to the quotient-remainder theorem, when an integer n is divided by 4, there are four possible remainders: 0, 1, 2, or 3. Therefore, n can be written in one of the following forms: $n = 4q$, $n = 4q+1$, $n = 4q+2$, or $n = 4q+3$, for some integer q.

We will consider each case for n^2:

Case 1: $n = 4q$ In this case, the square of n is:

$$\begin{aligned} n^2 &= (4q)^2 \\ n^2 &= 16q^2 \\ n^2 &= 4(4q^2) \end{aligned}$$

Let $k = 4q^2$. Since q is an integer, q^2 is an integer, and $4q^2$ is an integer. So, $n^2 = 4k$, which is of the form $4k$.

Case 2: $n = 4q+1$ In this case, the square of n is:

$$\begin{aligned} n^2 &= (4q+1)^2 \\ n^2 &= (4q)^2 + 2(4q)(1) + (1)^2 \\ n^2 &= 16q^2 + 8q + 1 \\ n^2 &= 4(4q^2 + 2q) + 1 \end{aligned}$$

Let $k = 4q^2 + 2q$. Since q is an integer, q^2 is an integer, $4q^2$ is an integer, $2q$ is an integer, and their sum $4q^2 + 2q$ is an integer. So, $n^2 = 4k + 1$, which is of the form $4k + 1$.

Case 3: $n = 4q+2$ In this case, the square of n is:

$$\begin{aligned} n^2 &= (4q+2)^2 \\ n^2 &= (4q)^2 + 2(4q)(2) + (2)^2 \\ n^2 &= 16q^2 + 16q + 4 \\ n^2 &= 4(4q^2 + 4q + 1) \end{aligned}$$

Let $k = 4q^2 + 4q + 1$. Since q is an integer, q^2 is an integer, $4q^2$ is an integer, $4q$ is an integer, and their sum $4q^2 + 4q + 1$ is an integer. So, $n^2 = 4k$, which is of the form $4k$.

Case 4: $n = 4q+3$ In this case, the square of n is:

$$\begin{aligned} n^2 &= (4q+3)^2 \\ n^2 &= (4q)^2 + 2(4q)(3) + (3)^2 \\ n^2 &= 16q^2 + 24q + 9 \\ n^2 &= 16q^2 + 24q + 8 + 1 \\ n^2 &= 4(4q^2 + 6q + 2) + 1 \end{aligned}$$

Let $k = 4q^2 + 6q + 2$. Since q is an integer, q^2 is an integer, $4q^2$ is an integer, $6q$ is an integer, and their sum $4q^2 + 6q + 2$ is an integer. So, $n^2 = 4k + 1$, which is of the form $4k + 1$.

In all possible cases, the square of an integer n is either of the form $4k$ or $4k+1$ for some integer k. Therefore, the square of any integer has the form $4k$ or $4k+1$ for some integer k.

Problem 38

H 38. For any integer n, $n^2 + 5$ is not divisible by 4.

We will prove this statement by considering the possible remainders when an integer n is divided by 4. According to the quotient-remainder theorem, for any integer n and divisor 4, there exist unique integers q and r such that $n = 4q + r$, where $0 \leq r < 4$. Thus, the possible values for r are 0, 1, 2, and 3.

We will analyze the value of $n^2 + 5$ for each of these cases.

Case 1: $n \equiv 0 \pmod 4$ This means n can be written in the form $n = 4q$ for some integer q. Then, $n^2 + 5 = (4q)^2 + 5$ $n^2 + 5 = 16q^2 + 5$

We want to check if $n^2 + 5$ is divisible by 4. $n^2 + 5 \pmod 4 = (16q^2 + 5) \pmod 4$ $n^2 + 5 \pmod 4 = (16q^2 \pmod 4 + 5 \pmod 4) \pmod 4$ Since $16q^2$ is a multiple of 16, it is also a multiple of 4. Thus, $16q^2 \pmod 4 = 0$. $n^2 + 5 \pmod 4 = (0 + 5 \pmod 4) \pmod 4$ $n^2 + 5 \pmod 4 = 1 \pmod 4$ Since the remainder is 1, $n^2 + 5$ is not divisible by 4 in this case.

Case 2: $n \equiv 1 \pmod 4$ This means n can be written in the form $n = 4q + 1$ for some integer q. Then, $n^2 + 5 = (4q + 1)^2 + 5$ $n^2 + 5 = (16q^2 + 8q + 1) + 5$ $n^2 + 5 = 16q^2 + 8q + 6$

We want to check if $n^2 + 5$ is divisible by 4. $n^2 + 5 \pmod 4 = (16q^2 + 8q + 6) \pmod 4$ $n^2 + 5 \pmod 4 = (16q^2 \pmod 4 + 8q \pmod 4 + 6 \pmod 4) \pmod 4$ Since $16q^2$ and $8q$ are multiples of 4, their remainders when divided by 4 are 0. $n^2 + 5 \pmod 4 = (0 + 0 + 6 \pmod 4) \pmod 4$ $n^2 + 5 \pmod 4 = 2 \pmod 4$ Since the remainder is 2, $n^2 + 5$ is not divisible by 4 in this case.

Case 3: $n \equiv 2 \pmod 4$ This means n can be written in the form $n = 4q + 2$ for some integer q. Then, $n^2 + 5 = (4q + 2)^2 + 5$ $n^2 + 5 = (16q^2 + 16q + 4) + 5$ $n^2 + 5 = 16q^2 + 16q + 9$

We want to check if $n^2 + 5$ is divisible by 4. $n^2 + 5 \pmod 4 = (16q^2 + 16q + 9) \pmod 4$ $n^2 + 5 \pmod 4 = (16q^2 \pmod 4 + 16q \pmod 4 + 9 \pmod 4) \pmod 4$ Since $16q^2$ and $16q$ are multiples of 4, their remainders when divided by 4 are 0. $n^2 + 5 \pmod 4 = (0 + 0 + 9 \pmod 4) \pmod 4$ $n^2 + 5 \pmod 4 = 1 \pmod 4$ Since the remainder is 1, $n^2 + 5$ is not divisible by 4 in this case.

Case 4: $n \equiv 3 \pmod 4$ This means n can be written in the form $n = 4q + 3$ for some integer q. Then, $n^2 + 5 = (4q + 3)^2 + 5$ $n^2 + 5 = (16q^2 + 24q + 9) + 5$ $n^2 + 5 = 16q^2 + 24q + 14$

We want to check if $n^2 + 5$ is divisible by 4. $n^2 + 5 \pmod 4 = (16q^2 + 24q + 14) \pmod 4$ $n^2 + 5 \pmod 4 = (16q^2 \pmod 4 + 24q \pmod 4 + 14 \pmod 4) \pmod 4$ Since $16q^2$ and $24q$ are multiples of 4, their remainders when divided by 4 are 0. $n^2 + 5 \pmod 4 = (0 + 0 + 14 \pmod 4) \pmod 4$ $n^2 + 5 \pmod 4 = 2 \pmod 4$ Since the remainder is 2, $n^2 + 5$ is not divisible by 4 in this case.

In all possible cases for the remainder of n when divided by 4, the remainder of $n^2 + 5$ when divided by 4 is either 1 or 2. Therefore, $n^2 + 5$ is never divisible by 4 for any integer n.

Problem 39

Problem 39 asks to prove that for any integer n, the sum of any four consecutive integers has the form $4k + 2$ for some integer k.

Let the first of the four consecutive integers be n. Then the four consecutive integers can be represented as n, $n + 1$, $n + 2$, and $n + 3$.

We need to find the sum of these four integers: Sum $= n + (n + 1) + (n + 2) + (n + 3)$

Combine the terms: Sum $= (n + n + n + n) + (1 + 2 + 3)$

Sum $= 4n + 6$

Now we want to express this sum in the form $4k + 2$. We can rewrite 6 as $4 + 2$: Sum $= 4n + 4 + 2$

Factor out 4 from the first two terms: Sum $= 4(n + 1) + 2$

Let $k = n + 1$. Since n is an integer, $n + 1$ is also an integer. Therefore, k is an integer.

Substituting k back into the expression for the sum: Sum $= 4k + 2$

Thus, the sum of any four consecutive integers can be written in the form $4k + 2$ for some integer k.

The final answer is $\boxed{4k + 2}$.

Problem 40

For any integer n, $n(n^2-1)(n+2)$ is divisible by 4.

We can rewrite the expression as $n(n-1)(n+1)(n+2)$. This is a product of four consecutive integers.

We need to show that this product is divisible by 4. We can consider the possible remainders of n when divided by 4.

Case 1: n is a multiple of 4. In this case, $n = 4k$ for some integer k. Then $n(n-1)(n+1)(n+2) = (4k)(4k-1)(4k+1)(4k+2)$. Since $4k$ is a factor, the entire product is divisible by 4.

Case 2: n has a remainder of 1 when divided by 4. In this case, $n = 4k+1$ for some integer k. Then $n(n-1)(n+1)(n+2) = (4k+1)((4k+1)-1)((4k+1)+1)((4k+1)+2) = (4k+1)(4k)(4k+2)(4k+3)$. Since $4k$ is a factor, the entire product is divisible by 4.

Case 3: n has a remainder of 2 when divided by 4. In this case, $n = 4k+2$ for some integer k. Then $n(n-1)(n+1)(n+2) = (4k+2)((4k+2)-1)((4k+2)+1)((4k+2)+2) = (4k+2)(4k+1)(4k+3)(4k+4)$. We can rewrite $4k+4$ as $4(k+1)$. So the expression becomes $(4k+2)(4k+1)(4k+3) \cdot 4(k+1)$. Since $4(k+1)$ is a factor, the entire product is divisible by 4.

Case 4: n has a remainder of 3 when divided by 4. In this case, $n = 4k+3$ for some integer k. Then $n(n-1)(n+1)(n+2) = (4k+3)((4k+3)-1)((4k+3)+1)((4k+3)+2) = (4k+3)(4k+2)(4k+4)(4k+5)$. We can rewrite $4k+4$ as $4(k+1)$. So the expression becomes $(4k+3)(4k+2) \cdot 4(k+1) \cdot (4k+5)$. Since $4(k+1)$ is a factor, the entire product is divisible by 4.

In all possible cases for n when divided by 4, the product $n(n^2-1)(n+2)$ is divisible by 4.

Alternatively, we can use the quotient-remainder theorem with divisor 2. Any integer n is either even or odd.

If n is even, then $n = 2k$ for some integer k. Then $n(n^2-1)(n+2) = (2k)((2k)^2-1)(2k+2) = (2k)(4k^2-1)(2(k+1)) = 4k(k+1)(4k^2-1)$. Since there is a factor of 4, the product is divisible by 4.

If n is odd, then $n = 2k+1$ for some integer k. Then $n^2-1 = (2k+1)^2-1 = (4k^2+4k+1)-1 = 4k^2+4k = 4k(k+1)$. So, $n(n^2-1)(n+2) = (2k+1)(4k(k+1))(2k+1+2) = (2k+1)(4k(k+1))(2k+3)$. Since there is a factor of 4, the product is divisible by 4.

Therefore, for any integer n, $n(n^2-1)(n+2)$ is divisible by 4.

Problem 41

41. For all integers m, $m^2 = 5k$, or $m^2 = 5k + 1$, or $m^2 = 5k + 4$ for some integer k.

We will use the quotient-remainder theorem. For any integer m, when divided by 5, there are five possible remainders: 0, 1, 2, 3, or 4. Thus, we can write m in one of the following forms: 1. $m = 5q$ for some integer q. 2. $m = 5q + 1$ for some integer q. 3. $m = 5q + 2$ for some integer q. 4. $m = 5q + 3$ for some integer q. 5. $m = 5q + 4$ for some integer q.

We will consider the square of m in each of these cases.

Case 1: $m = 5q$

$$\begin{aligned} m^2 &= (5q)^2 \\ &= 25q^2 \\ &= 5(5q^2) \end{aligned}$$

Let $k = 5q^2$. Since q is an integer, q^2 is an integer, and $5q^2$ is an integer. Therefore, $m^2 = 5k$ for some integer k.

Case 2: $m = 5q + 1$

$$\begin{aligned} m^2 &= (5q + 1)^2 \\ &= (5q)^2 + 2(5q)(1) + 1^2 \\ &= 25q^2 + 10q + 1 \\ &= 5(5q^2 + 2q) + 1 \end{aligned}$$

Let $k = 5q^2 + 2q$. Since q is an integer, q^2 is an integer, $2q$ is an integer, and $5q^2 + 2q$ is an integer. Therefore, $m^2 = 5k + 1$ for some integer k.

Case 3: $m = 5q + 2$

$$\begin{aligned} m^2 &= (5q + 2)^2 \\ &= (5q)^2 + 2(5q)(2) + 2^2 \\ &= 25q^2 + 20q + 4 \\ &= 5(5q^2 + 4q) + 4 \end{aligned}$$

Let $k = 5q^2 + 4q$. Since q is an integer, q^2 is an integer, $4q$ is an integer, and $5q^2 + 4q$ is an integer. Therefore, $m^2 = 5k + 4$ for some integer k.

Case 4: $m = 5q + 3$

$$\begin{aligned} m^2 &= (5q + 3)^2 \\ &= (5q)^2 + 2(5q)(3) + 3^2 \\ &= 25q^2 + 30q + 9 \\ &= 25q^2 + 30q + 5 + 4 \\ &= 5(5q^2 + 6q + 1) + 4 \end{aligned}$$

Let $k = 5q^2 + 6q + 1$. Since q is an integer, q^2 is an integer, $6q$ is an integer, and $5q^2 + 6q + 1$ is an integer. Therefore, $m^2 = 5k + 4$ for some integer k.

Case 5: $m = 5q + 4$

$$\begin{aligned} m^2 &= (5q+4)^2 \\ &= (5q)^2 + 2(5q)(4) + 4^2 \\ &= 25q^2 + 40q + 16 \\ &= 25q^2 + 40q + 15 + 1 \\ &= 5(5q^2 + 8q + 3) + 1 \end{aligned}$$

Let $k = 5q^2 + 8q + 3$. Since q is an integer, q^2 is an integer, $8q$ is an integer, and $5q^2 + 8q + 3$ is an integer. Therefore, $m^2 = 5k + 1$ for some integer k.

In summary, for any integer m, its square m^2 must be of the form $5k$, $5k + 1$, or $5k + 4$ for some integer k.

Problem 42

42. Every prime number except 2 and 3 has the form 6q + 1 or 6q + 5 for some integer q.

We will use the Quotient-Remainder Theorem. For any integer p, when divided by 6, there are six possible remainders: 0, 1, 2, 3, 4, or 5. Thus, any integer p can be written in one of the following forms for some integer q:

$$p = 6q$$
$$p = 6q + 1$$
$$p = 6q + 2$$
$$p = 6q + 3$$
$$p = 6q + 4$$
$$p = 6q + 5$$

We need to consider prime numbers p greater than 3.

Case 1: $p = 6q$. If $p = 6q$, then p is divisible by 6, and therefore by 2 and 3. If $q = 1$, $p = 6$, which is not prime. If $q > 1$, p has factors other than 1 and itself, so it cannot be prime. If $q = 0$, $p = 0$, which is not prime. If $q < 0$, then p is negative, and prime numbers are positive.

Case 2: $p = 6q + 2$. If $p = 6q + 2$, we can factor out a 2: $p = 2(3q + 1)$. This means p is divisible by 2. If $3q + 1 = 1$, then $q = 0$, which gives $p = 2$. The number 2 is prime. If $3q + 1 \neq 1$ (i.e., $q \neq 0$), then p has a factor of 2 and another factor $(3q + 1)$, so p is not prime if $p > 2$.

Case 3: $p = 6q + 3$. If $p = 6q + 3$, we can factor out a 3: $p = 3(2q + 1)$. This means p is divisible by 3. If $2q + 1 = 1$, then $q = 0$, which gives $p = 3$. The number 3 is prime. If $2q + 1 \neq 1$ (i.e., $q \neq 0$), then p has a factor of 3 and another factor $(2q + 1)$, so p is not prime if $p > 3$.

Case 4: $p = 6q + 4$. If $p = 6q + 4$, we can factor out a 2: $p = 2(3q + 2)$. This means p is divisible by 2. If $3q + 2 = 1$, then $3q = -1$, which has no integer solution for q. Thus, p always has a factor of 2 and another factor $(3q + 2)$, so p is not prime.

The remaining forms are $p = 6q + 1$ and $p = 6q + 5$.

If a prime number p is not 2 or 3, it cannot be of the form $6q$, $6q + 2$, $6q + 3$, or $6q + 4$. Therefore, any prime number p except 2 and 3 must be of the form $6q + 1$ or $6q + 5$ for some integer q.

For example: If $q = 1$, $6q + 1 = 7$ (prime) and $6q + 5 = 11$ (prime). If $q = 2$, $6q + 1 = 13$ (prime) and $6q + 5 = 17$ (prime). If $q = 3$, $6q + 1 = 19$ (prime) and $6q + 5 = 23$ (prime). If $q = 4$, $6q + 1 = 25$ (not prime) and $6q + 5 = 29$ (prime). This demonstrates that not all numbers of the form $6q + 1$ or $6q + 5$ are prime, but all primes greater than 3 must be of these forms.

Problem 43

We want to prove that if n is an odd integer, then $n^4 \bmod 16 = 1$.

Since n is an odd integer, we can write n in one of the following forms based on the quotient-remainder theorem with divisor $d = 2$: $n = 2k + 1$ for some integer k.

We need to consider the possible values of k modulo 8. This is because when we raise n to the fourth power, the resulting number will be related to powers of 2 up to $2^4 = 16$. Analyzing k modulo 8 will help us understand the behavior of n^4 modulo 16.

Case 1: $k = 8m$ for some integer m. Then $n = 2(8m) + 1 = 16m + 1$.

$$\begin{aligned}
n^4 &= (16m+1)^4 \\
&= \binom{4}{0}(16m)^4(1)^0 + \binom{4}{1}(16m)^3(1)^1 + \binom{4}{2}(16m)^2(1)^2 + \binom{4}{3}(16m)^1(1)^3 + \binom{4}{4}(16m)^0(1)^4 \\
&= 1(16m)^4 + 4(16m)^3 + 6(16m)^2 + 4(16m) + 1 \\
&= 16^4m^4 + 4 \cdot 16^3m^3 + 6 \cdot 16^2m^2 + 4 \cdot 16m + 1
\end{aligned}$$

We can see that every term except the last term is a multiple of 16. Thus, $n^4 = 16(\dots) + 1$. Therefore, $n^4 \bmod 16 = 1$.

Case 2: $k = 8m + 1$ for some integer m. Then $n = 2(8m + 1) + 1 = 16m + 2 + 1 = 16m + 3$.

$$\begin{aligned}
n^4 &= (16m+3)^4 \\
&= \binom{4}{0}(16m)^4(3)^0 + \binom{4}{1}(16m)^3(3)^1 + \binom{4}{2}(16m)^2(3)^2 + \binom{4}{3}(16m)^1(3)^3 + \binom{4}{4}(16m)^0(3)^4 \\
&= 1(16m)^4 + 4(16m)^3 \cdot 3 + 6(16m)^2 \cdot 9 + 4(16m) \cdot 27 + 1 \cdot 81 \\
&= 16^4m^4 + 12 \cdot 16^3m^3 + 54 \cdot 16^2m^2 + 108 \cdot 16m + 81
\end{aligned}$$

We can rewrite 81 as $5 \times 16 + 1$. So, $n^4 = 16(\dots) + 81 = 16(\dots) + 5 \times 16 + 1 = 16(\dots) + 1$. Therefore, $n^4 \bmod 16 = 1$.

Case 3: $k = 8m + 2$ for some integer m. Then $n = 2(8m + 2) + 1 = 16m + 4 + 1 = 16m + 5$.

$$\begin{aligned}
n^4 &= (16m+5)^4 \\
&= \binom{4}{0}(16m)^4(5)^0 + \binom{4}{1}(16m)^3(5)^1 + \binom{4}{2}(16m)^2(5)^2 + \binom{4}{3}(16m)^1(5)^3 + \binom{4}{4}(16m)^0(5)^4 \\
&= 1(16m)^4 + 4(16m)^3 \cdot 5 + 6(16m)^2 \cdot 25 + 4(16m) \cdot 125 + 1 \cdot 625 \\
&= 16^4m^4 + 20 \cdot 16^3m^3 + 150 \cdot 16^2m^2 + 500 \cdot 16m + 625
\end{aligned}$$

We can rewrite 625 as $39 \times 16 + 1$. So, $n^4 = 16(\dots) + 625 = 16(\dots) + 39 \times 16 + 1 = 16(\dots) + 1$. Therefore, $n^4 \bmod 16 = 1$.

Case 4: $k = 8m + 3$ for some integer m. Then $n = 2(8m + 3) + 1 = 16m + 6 + 1 = 16m + 7$.

$$\begin{aligned}
n^4 &= (16m+7)^4 \\
&= \binom{4}{0}(16m)^4(7)^0 + \binom{4}{1}(16m)^3(7)^1 + \binom{4}{2}(16m)^2(7)^2 + \binom{4}{3}(16m)^1(7)^3 + \binom{4}{4}(16m)^0(7)^4 \\
&= 1(16m)^4 + 4(16m)^3 \cdot 7 + 6(16m)^2 \cdot 49 + 4(16m) \cdot 343 + 1 \cdot 2401 \\
&= 16^4m^4 + 28 \cdot 16^3m^3 + 294 \cdot 16^2m^2 + 1372 \cdot 16m + 2401
\end{aligned}$$

We can rewrite 2401 as $150 \times 16 + 1$. So, $n^4 = 16(\dots) + 2401 = 16(\dots) + 150 \times 16 + 1 = 16(\dots) + 1$.

Therefore, $n^4 \bmod 16 = 1$.

Case 5: $k = 8m + 4$ for some integer m. Then $n = 2(8m + 4) + 1 = 16m + 8 + 1 = 16m + 9$.

$$\begin{aligned}
n^4 &= (16m + 9)^4 \\
&= \binom{4}{0}(16m)^4(9)^0 + \binom{4}{1}(16m)^3(9)^1 + \binom{4}{2}(16m)^2(9)^2 + \binom{4}{3}(16m)^1(9)^3 + \binom{4}{4}(16m)^0(9)^4 \\
&= 1(16m)^4 + 4(16m)^3 \cdot 9 + 6(16m)^2 \cdot 81 + 4(16m) \cdot 729 + 1 \cdot 6561 \\
&= 16^4m^4 + 36 \cdot 16^3m^3 + 486 \cdot 16^2m^2 + 2916 \cdot 16m + 6561
\end{aligned}$$

We can rewrite 6561 as $410 \times 16 + 1$. So, $n^4 = 16(\dots) + 6561 = 16(\dots) + 410 \times 16 + 1 = 16(\dots) + 1$. Therefore, $n^4 \bmod 16 = 1$.

Case 6: $k = 8m + 5$ for some integer m. Then $n = 2(8m + 5) + 1 = 16m + 10 + 1 = 16m + 11$.

$$\begin{aligned}
n^4 &= (16m + 11)^4 \\
&= \binom{4}{0}(16m)^4(11)^0 + \binom{4}{1}(16m)^3(11)^1 + \binom{4}{2}(16m)^2(11)^2 + \binom{4}{3}(16m)^1(11)^3 + \binom{4}{4}(16m)^0(11)^4 \\
&= 1(16m)^4 + 4(16m)^3 \cdot 11 + 6(16m)^2 \cdot 121 + 4(16m) \cdot 1331 + 1 \cdot 14641 \\
&= 16^4m^4 + 44 \cdot 16^3m^3 + 726 \cdot 16^2m^2 + 5324 \cdot 16m + 14641
\end{aligned}$$

We can rewrite 14641 as $915 \times 16 + 1$. So, $n^4 = 16(\dots) + 14641 = 16(\dots) + 915 \times 16 + 1 = 16(\dots) + 1$. Therefore, $n^4 \bmod 16 = 1$.

Case 7: $k = 8m + 6$ for some integer m. Then $n = 2(8m + 6) + 1 = 16m + 12 + 1 = 16m + 13$.

$$\begin{aligned}
n^4 &= (16m + 13)^4 \\
&= \binom{4}{0}(16m)^4(13)^0 + \binom{4}{1}(16m)^3(13)^1 + \binom{4}{2}(16m)^2(13)^2 + \binom{4}{3}(16m)^1(13)^3 + \binom{4}{4}(16m)^0(13)^4 \\
&= 1(16m)^4 + 4(16m)^3 \cdot 13 + 6(16m)^2 \cdot 169 + 4(16m) \cdot 2197 + 1 \cdot 28561 \\
&= 16^4m^4 + 52 \cdot 16^3m^3 + 1014 \cdot 16^2m^2 + 8788 \cdot 16m + 28561
\end{aligned}$$

We can rewrite 28561 as $1785 \times 16 + 1$. So, $n^4 = 16(\dots) + 28561 = 16(\dots) + 1785 \times 16 + 1 = 16(\dots) + 1$. Therefore, $n^4 \bmod 16 = 1$.

Case 8: $k = 8m + 7$ for some integer m. Then $n = 2(8m + 7) + 1 = 16m + 14 + 1 = 16m + 15$.

$$\begin{aligned}
n^4 &= (16m + 15)^4 \\
&= \binom{4}{0}(16m)^4(15)^0 + \binom{4}{1}(16m)^3(15)^1 + \binom{4}{2}(16m)^2(15)^2 + \binom{4}{3}(16m)^1(15)^3 + \binom{4}{4}(16m)^0(15)^4 \\
&= 1(16m)^4 + 4(16m)^3 \cdot 15 + 6(16m)^2 \cdot 225 + 4(16m) \cdot 3375 + 1 \cdot 50625 \\
&= 16^4m^4 + 60 \cdot 16^3m^3 + 1350 \cdot 16^2m^2 + 13500 \cdot 16m + 50625
\end{aligned}$$

We can rewrite 50625 as $3164 \times 16 + 1$. So, $n^4 = 16(\dots) + 50625 = 16(\dots) + 3164 \times 16 + 1 = 16(\dots) + 1$. Therefore, $n^4 \bmod 16 = 1$.

Alternatively, we can observe that an odd integer n can be written in the form $n = 2k + 1$. We consider $n \bmod 8$. An odd integer can be written as $n \equiv 1, 3, 5, 7 \pmod 8$.

If $n \equiv 1 \pmod 8$, then $n = 8m + 1$ for some integer m. $n^2 = (8m + 1)^2 = 64m^2 + 16m + 1 = 16(4m^2 + m) + 1 \equiv 1 \pmod{16}$. $n^4 = (n^2)^2 \equiv 1^2 \equiv 1 \pmod{16}$.

If $n \equiv 3 \pmod 8$, then $n = 8m + 3$ for some integer m. $n^2 = (8m + 3)^2 = 64m^2 + 48m + 9 = 16(4m^2 + 3m) + 9 \equiv 9 \pmod{16}$. $n^4 = (n^2)^2 \equiv 9^2 = 81 = 5 \times 16 + 1 \equiv 1 \pmod{16}$.

If $n \equiv 5 \pmod 8$, then $n = 8m + 5$ for some integer m. $n^2 = (8m+5)^2 = 64m^2 + 80m + 25 = 16(4m^2 + 5m) + 25 = 16(4m^2 + 5m) + 16 + 9 = 16(4m^2 + 5m + 1) + 9 \equiv 9 \pmod{16}$. $n^4 = (n^2)^2 \equiv 9^2 = 81 = 5 \times 16 + 1 \equiv 1 \pmod{16}$.

If $n \equiv 7 \pmod 8$, then $n = 8m + 7$ for some integer m. $n^2 = (8m+7)^2 = 64m^2 + 112m + 49 = 16(4m^2+7m)+49 = 16(4m^2+7m)+3\times16+1 = 16(4m^2+7m+3)+1 \equiv 1 \pmod{16}$. $n^4 = (n^2)^2 \equiv 1^2 \equiv 1 \pmod{16}$.

In all cases for an odd integer n, $n^4 \equiv 1 \pmod{16}$.

The final answer is $\boxed{n^4 \bmod 16 = 1}$.

Problem 44

Problem 44: For all real numbers x and y, $|x||y| = |xy|$.

Proof:

We will consider several cases based on the signs of x and y.

Case 1: $x \geq 0$ and $y \geq 0$. In this case, $|x| = x$ and $|y| = y$. Also, $xy \geq 0$, so $|xy| = xy$. Therefore, $|x||y| = x \cdot y = xy = |xy|$.

$$\begin{aligned} |x||y| &= x \cdot y \\ &= xy \\ &= |xy| \end{aligned}$$

Case 2: $x < 0$ and $y < 0$. In this case, $|x| = -x$ and $|y| = -y$. Also, $xy > 0$ (since the product of two negative numbers is positive), so $|xy| = xy$. Therefore, $|x||y| = (-x)(-y) = xy = |xy|$.

$$\begin{aligned} |x||y| &= (-x)(-y) \\ &= xy \\ &= |xy| \end{aligned}$$

Case 3: $x \geq 0$ and $y < 0$. In this case, $|x| = x$ and $|y| = -y$. Also, $xy \leq 0$ (since the product of a non-negative number and a negative number is non-positive), so $|xy| = -(xy)$. Therefore, $|x||y| = x(-y) = -xy = |xy|$.

$$\begin{aligned} |x||y| &= x(-y) \\ &= -xy \\ &= |xy| \end{aligned}$$

Case 4: $x < 0$ and $y \geq 0$. In this case, $|x| = -x$ and $|y| = y$. Also, $xy \leq 0$ (since the product of a negative number and a non-negative number is non-positive), so $|xy| = -(xy)$. Therefore, $|x||y| = (-x)y = -xy = |xy|$.

$$\begin{aligned} |x||y| &= (-x)y \\ &= -xy \\ &= |xy| \end{aligned}$$

Since the equality $|x||y| = |xy|$ holds in all possible cases, the statement is proven.

Problem 45

For all real numbers r and c with $c > 0$, if $-c \leq r \leq c$, then $|r| \leq c$.

To prove this statement, we need to show that if $-c \leq r \leq c$ is true, then $|r| \leq c$ must also be true.

We can consider two cases for the real number r:

Case 1: $r \geq 0$. In this case, by the definition of the absolute value, $|r| = r$. We are given that $r \leq c$. Since $|r| = r$, we can substitute r with $|r|$ in the inequality $r \leq c$. Thus, we have $|r| \leq c$.

Case 2: $r < 0$. In this case, by the definition of the absolute value, $|r| = -r$. We are given that $-c \leq r$. Multiplying both sides of the inequality $-c \leq r$ by -1, we reverse the inequality sign: $(-1)(-c) \geq (-1)r$ $c \geq -r$ Since $|r| = -r$, we can substitute $-r$ with $|r|$ in the inequality $c \geq -r$. Thus, we have $c \geq |r|$, which can also be written as $|r| \leq c$.

In both cases ($r \geq 0$ and $r < 0$), we have shown that if $-c \leq r \leq c$, then $|r| \leq c$.

Therefore, the statement is proven.

Problem 46

46. For all real numbers r and c with c ¿ 0, if r c, then -c r c.

Proof:

We are given that r and c are real numbers with $c > 0$, and $r \leq c$. We need to prove that $-c \leq r \leq c$.

The statement $r \leq c$ is already given, so we only need to prove that $-c \leq r$.

We know that c is a positive real number, which means $c > 0$. Multiplying both sides of the inequality $c > 0$ by -1, and reversing the inequality sign, we get:

$$-1 \cdot c < -1 \cdot 0$$

$$-c < 0$$

We are given that $r \leq c$. Since c is a real number, we can add $-c$ to both sides of the inequality $r \leq c$:

$$r + (-c) \leq c + (-c)$$

$$r - c \leq 0$$

We have established that $-c < 0$. We also have $r - c \leq 0$.

Consider the relationship between r and $-c$. We know that $r \leq c$. We also know that $-c < 0$.

Let's consider the inequality $-c \leq r$. This is equivalent to showing that $r - (-c) \geq 0$, which means $r + c \geq 0$.

We are given $r \leq c$. This means that $r - c \leq 0$.

Let's consider the case where r is positive. If $r > 0$ and $r \leq c$, then we also have $0 < r \leq c$. Since $c > 0$, it follows that $-c < 0 < r \leq c$. Thus, $-c \leq r \leq c$.

Let's consider the case where r is zero. If $r = 0$ and $r \leq c$, then $0 \leq c$. Since we are given $c > 0$, this is true. In this case, $-c \leq 0 \leq c$, which means $-c \leq r \leq c$.

Let's consider the case where r is negative. If $r < 0$ and $r \leq c$. We know that $-c < 0$. We want to show that $-c \leq r$.

Consider the property that for any real numbers x and y, if $x \leq y$, then for any z, $x + z \leq y + z$.

We are given $r \leq c$. Since $c > 0$, we know that $-c < 0$.

Let's add $-c$ to both sides of $r \leq c$:

$$r + (-c) \leq c + (-c)$$

$$r - c \leq 0$$

Now, let's consider the inequality $-c \leq r$. This is equivalent to $0 \leq r + c$.

We have $r \leq c$. Consider the term $r + c$.

Let's use the given inequality $r \leq c$. Subtract c from both sides:

$$r - c \leq c - c$$

$$r - c \leq 0$$

We also know that $c > 0$. This implies that $-c < 0$.

We want to show $-c \leq r$. This is equivalent to showing that $0 \leq r - (-c)$, or $0 \leq r + c$.

Let's consider the given condition $r \leq c$. If we take the negative of both sides and reverse the inequality, we would need more information.

Let's go back to $-c < 0$. We are given $r \leq c$.

Consider the expression $r + c$. We know $r \leq c$. If we add c to both sides of the inequality $r \leq c$:

$$r + c \leq c + c$$

$$r + c \leq 2c$$

This does not directly help us show $r + c \geq 0$.

Let's consider the statement $r \leq c$. We want to show $-c \leq r$. This means that r must be greater than or equal to $-c$.

Since $c > 0$, we know that $-c$ is a negative number. If $r \leq c$, and c is positive, then r could be positive, zero, or negative.

Let's consider the inequality $-c \leq r$. We are given $r \leq c$.

If $r \leq c$, and $c > 0$. We want to prove $-c \leq r$.

Let's rewrite the statement we want to prove as two separate inequalities: 1. $r \leq c$ (This is given) 2. $-c \leq r$

We need to prove $-c \leq r$. We know that $c > 0$. Therefore, $-c < 0$.

Consider the inequality $r \leq c$. If we add $-c$ to both sides, we get $r - c \leq c - c$, which is $r - c \leq 0$.

Let's consider the inequality $-c \leq r$. This is equivalent to $0 \leq r + c$.

We have $r \leq c$. Consider the expression $r + c$.

Let's use the fact that $c > 0$. This implies that $0 < c$.

We are given $r \leq c$. We want to show $-c \leq r$.

Consider the number line. If r is to the left of or at c. We want to show that $-c$ is to the left of or at r.

Since $c > 0$, we know that $-c$ is to the left of 0. Since $r \leq c$, r is to the left of or at c.

Let's subtract r from both sides of $r \leq c$:

$$r - r \leq c - r$$

$$0 \leq c - r$$

This means $c - r \geq 0$.

We want to show $-c \leq r$. This is equivalent to $0 \leq r + c$.

Consider the given $r \leq c$. Add c to both sides.

$$r + c \leq c + c$$

$$r + c \leq 2c$$

This is not what we want.

Let's consider the property of real numbers. If $a \leq b$ and $d \leq e$, then $a + d \leq b + e$.

We are given $r \leq c$. We know that $-c < 0$.

We need to prove $-c \leq r$. This is equivalent to proving $r - (-c) \geq 0$, which is $r + c \geq 0$.

Consider the inequality $r \leq c$. We want to show $-c \leq r$.

Let's consider the negation of what we want to show. Assume $-c > r$. This means $r < -c$. Since $c > 0$, we have $-c < 0$. So $r < -c < 0$. If $r < -c$, then adding c to both sides gives:

$$r + c < -c + c$$

$$r + c < 0$$

However, we are given $r \leq c$. If we add c to both sides:

$$r + c \leq c + c$$

$$r + c \leq 2c$$

Let's consider the implication: If $r \leq c$, then $-c \leq r$. We know $c > 0$.

Consider the number line. If r is at or to the left of c. We know $-c$ is to the left of 0.

Let's subtract r from c: $c - r \geq 0$. We want to show $r - (-c) \geq 0$, which is $r + c \geq 0$.

We have $r \leq c$. We know $c > 0$.

Consider the case where r is positive. If $0 < r \leq c$. Since $c > 0$, we have $-c < 0 < r \leq c$. Thus $-c \leq r \leq c$.

Consider the case where r is zero. If $r = 0$ and $r \leq c$. Then $0 \leq c$. Since $c > 0$, this is true. We have $-c \leq 0 \leq c$, so $-c \leq r \leq c$.

Consider the case where r is negative. If $r < 0$ and $r \leq c$. We know that $-c < 0$. We want to show $-c \leq r$.

Let's consider the property: if $x \leq y$, then $-y \leq -x$. We are given $r \leq c$. Multiplying by -1 and reversing the inequality:

$$-1 \cdot r \geq -1 \cdot c$$

$$-r \geq -c$$

This means $-c \leq -r$.

We need to show $-c \leq r$.

Let's use the fact that $c > 0$. This implies $0 < c$. Also, $-c < 0$.

We are given $r \leq c$. This means that the distance from r to c is non-negative, i.e., $c - r \geq 0$.

We want to show $-c \leq r$. This is equivalent to $0 \leq r + c$.

Let's consider the inequality $r \leq c$. If we add c to both sides of $-c \leq r$, we get $-c + c \leq r + c$, which is $0 \leq r + c$. This is what we want to prove.

We have $r \leq c$. We need to show $-c \leq r$.

Let's consider the sum $r + c$. We know $r \leq c$. If we add c to both sides, we get $r + c \leq 2c$. If we add $-r$ to both sides, we get $0 \leq c - r$.

Let's consider the absolute value definition. If $r \leq c$, where $c > 0$. This means that r is on the interval $(-\infty, c]$.

We want to show $-c \leq r \leq c$. Since $r \leq c$ is given, we only need to show $-c \leq r$.

Since $c > 0$, we have $-c < 0$.

Consider the inequality $r \leq c$. If we add $-c$ to both sides:

$$r + (-c) \leq c + (-c)$$

$$r - c \leq 0$$

We want to show $-c \leq r$. This is equivalent to showing that $r - (-c) \geq 0$, which is $r + c \geq 0$.

Let's re-examine the statement: For all real numbers r and c with $c > 0$, if $r \leq c$, then $-c \leq r \leq c$.

The statement $-c \leq r \leq c$ is a compound inequality, meaning $-c \leq r$ AND $r \leq c$. We are given $r \leq c$. So, we only need to prove $-c \leq r$.

Since $c > 0$, we can multiply the inequality $c > 0$ by -1 to get $-c < 0$. We are given $r \leq c$.

Consider the value of r. If r is greater than or equal to 0, and $r \leq c$, then since $-c < 0 < r$, we have $-c < r$. So $-c \leq r$. If r is less than 0, and $r \leq c$. We need to show $-c \leq r$.

Let's consider the number line. We have $-c < 0$. We have $r \leq c$.

If $r < -c$, then since $-c < 0$, it implies $r < 0$. If $r < -c$, then $r + c < 0$.

We are given $r \leq c$. We want to show $-c \leq r$.

Let's consider the negation of $-c \leq r$, which is $-c > r$, or $r < -c$. If $r < -c$, since $c > 0$, we have $-c < 0$, so $r < -c < 0$. If $r < -c$, then $r + c < 0$.

However, we are given $r \leq c$. Adding c to both sides: $r + c \leq 2c$.

Consider the definition of $|r| \leq c$. This means $-c \leq r \leq c$. The problem statement is essentially saying that if $r \leq c$ (where $c > 0$), then the absolute value of r is less than or equal to c. This is not generally true. For example, if $c = 2$ and $r = 3$, then $r \nleq c$. If $c = 2$ and $r = -3$, then $r \leq c$ is true, but $-c \leq r \leq c$ is $-2 \leq -3 \leq 2$, which is false.

There might be a misunderstanding of the problem statement or a typo.

Let's assume the question meant to ask: Prove that for all real numbers r and c with $c > 0$, if $|r| \leq c$, then $-c \leq r \leq c$. This is the definition of absolute value.

However, the problem is stated as: "For all real numbers x and y, —x——y— = —xy—." and "For all real numbers r and c with c ¿ 0, if -c r c, then —r— c." and "For all real numbers r and c with c ¿ 0, if r c, then -c r c."

Let's re-read the problem to ensure no misinterpretation. The problem is question 46.

The previous problems are: 44. For all real numbers x and y, —x——y— = —xy—. (This is a property of absolute value) 45. For all real numbers r and c with c ¿ 0, if -c r c, then —r— c. (This is the definition of absolute value) 46. For all real numbers r and c with c ¿ 0, if r c, then -c r c.

Let's assume the problem statement is correct as given. We need to prove that if $r \leq c$ and $c > 0$, then $-c \leq r$ and $r \leq c$. The part $r \leq c$ is given. We only need to prove $-c \leq r$.

Let's consider a counterexample to see if the statement is false. Let $c = 5$. Since $c > 0$, this is valid. Let $r = 6$. Then $r \nleq c$. So this choice is not allowed by the hypothesis. Let $r = 4$. Then $r \leq c$ is $4 \leq 5$, which is true. We need to check if $-c \leq r \leq c$. This is $-5 \leq 4 \leq 5$. This is true.

Let $r = -6$. Then $r \leq c$ is $-6 \leq 5$, which is true. We need to check if $-c \leq r \leq c$. This is $-5 \leq -6 \leq 5$. This is false because $-5 \nleq -6$.

Therefore, the statement "For all real numbers r and c with c ¿ 0, if r c, then -c r c" is false.

There must be a misunderstanding or a typo in the problem statement. Based on the context of absolute value properties, it is highly probable that the condition should be $|r| \leq c$ or $-c \leq r \leq c$.

Assuming there is a typo and the statement meant to be: **46'. For all real numbers r and c with c ¿ 0, if —r— c, then -c r c.**

Proof of 46': We are given that r and c are real numbers with $c > 0$, and $|r| \leq c$. The definition of the absolute value $|r|$ is: $|r| = r$ if $r \geq 0$ $|r| = -r$ if $r < 0$

We need to show that $-c \leq r \leq c$. This consists of two inequalities: $r \leq c$ and $-c \leq r$.

Case 1: $r \geq 0$. In this case, $|r| = r$. The given condition $|r| \leq c$ becomes $r \leq c$. Since $r \geq 0$ and $c > 0$, it follows that $0 \leq r \leq c$. This implies $r \leq c$. Also, since $c > 0$, we have $-c < 0$. Since $r \geq 0$, we have $-c < 0 \leq r$, which means $-c \leq r$. Thus, in this case, $-c \leq r \leq c$.

Case 2: $r < 0$. In this case, $|r| = -r$. The given condition $|r| \leq c$ becomes $-r \leq c$. Multiplying both sides by -1 and reversing the inequality sign, we get:

$$-1 \cdot (-r) \geq -1 \cdot c$$

$$r \geq -c$$

So, we have $-c \leq r$. Also, since $r < 0$ and $c > 0$, it follows that $r < 0 < c$. This implies $r \leq c$. Thus, in this case, $-c \leq r \leq c$.

In both cases, we have shown that $-c \leq r \leq c$.

If the problem statement is precisely as written (46): As shown with the counterexample $c = 5, r = -6$, the statement "For all real numbers r and c with c ¿ 0, if r c, then -c r c" is false.

If we are forced to provide a "proof" for the statement as written, we would have to show that the premise is insufficient to guarantee the conclusion.

Let's assume the question is asking to prove the statement and that it is indeed true as written. We are given $r \leq c$ and $c > 0$. We need to prove $-c \leq r$.

Since $c > 0$, we have $0 < c$. Multiplying by -1, we get $-c < 0$.

We are given $r \leq c$. We want to show $-c \leq r$.

Consider the inequality $r \leq c$. Add $-c$ to both sides:

$$r + (-c) \leq c + (-c)$$

$$r - c \leq 0$$

We want to show $-c \leq r$. This is equivalent to $0 \leq r - (-c)$, or $0 \leq r + c$.

Let's consider the properties of inequalities. If $a \leq b$, then for any x, $a + x \leq b + x$.

We are given $r \leq c$. We want to show $-c \leq r$.

Consider the value of $r + c$. We know $r \leq c$. If we add c to both sides: $r + c \leq 2c$.

If $r \leq c$, it means r is to the left of or at c on the number line. We know $-c$ is to the left of 0.

If r is such that $r < -c$, then r is to the left of $-c$. Since $-c < 0$, then r is also to the left of 0. If $r < -c$, then $r + c < 0$.

However, if $r \leq c$ and $c > 0$, r can be negative and less than $-c$. For example, $c = 5$, $r = -10$. Here $r \leq c$ is $-10 \leq 5$, which is true. But $-c \leq r \leq c$ is $-5 \leq -10 \leq 5$, which is false.

Given the context of the preceding problems (44 and 45) which deal with absolute values and their definitions, it is highly probable that question 46 contains a typo and should have been "$|r| \leq c$". If not, the statement is false. Since this is a problem from a textbook section on proofs, and false statements are not typically presented as theorems to be proven without explicit mention, we proceed with the assumption of a typo and prove the corrected statement.

Assuming the statement to be proven is: **For all real numbers r and c with c ¿ 0, if —r— c, then -c r c.**

Proof:

We are given that r and c are real numbers with $c > 0$, and $|r| \leq c$. We need to prove that $-c \leq r \leq c$. This compound inequality means we need to prove two separate inequalities: 1. $r \leq c$ 2. $-c \leq r$

Proof of $r \leq c$: Since $|r| \leq c$ is given, and $c > 0$, we know that c is a positive value. The definition of absolute value is $|r| \geq 0$. Since $|r| \leq c$, and $c > 0$, the value of $|r|$ is bounded by c. If $r \geq 0$, then $|r| = r$. The condition $|r| \leq c$ becomes $r \leq c$. If $r < 0$, then $|r| = -r$. The condition $|r| \leq c$ becomes $-r \leq c$. Multiplying by -1 and reversing the inequality gives $r \geq -c$. However, we also know that if $r < 0$ and $c > 0$, then $r < 0 < c$, which implies $r \leq c$.

A more direct approach: Since $|r| \leq c$, by the definition of absolute value, this means that r is between $-c$ and c, inclusive.

$$-c \leq r \leq c$$

This directly implies that $r \leq c$.

Proof of $-c \leq r$: Again, since $|r| \leq c$ is given, by the definition of absolute value:

$$-c \leq r \leq c$$

This compound inequality implies that $-c \leq r$.

Therefore, combining both parts, if $|r| \leq c$ and $c > 0$, then $-c \leq r \leq c$.

If the statement is precisely as written: The statement "For all real numbers r and c with c ¿ 0, if r c, then -c r c" is false.

A proof of falsity involves a counterexample. Let $c = 5$. Since $c > 0$, this satisfies the condition $c > 0$. Let $r = -6$. We check the hypothesis: $r \leq c$. Is $-6 \leq 5$? Yes, this is true.

Now we check the conclusion: $-c \leq r \leq c$. This means $-5 \leq -6 \leq 5$. The inequality $-5 \leq -6$ is false. Since the conclusion is false while the hypothesis is true for this specific choice of r and c, the statement is false.

Problem 47

Here's the solution to problem 47, which involves understanding matrix storage in computer memory.

47. A matrix M has 3 rows and 4 columns.

$$M = \begin{pmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{pmatrix}$$

The 12 entries in the matrix are to be stored in row major form in locations 7,609 to 7,620 in a computer's memory. This means that the entries in the first row (reading left to right) are stored first, then the entries in the second row, and finally the entries in the third row.

a. Which location will a_{22} be stored in?

In row major order, the elements are stored row by row. The matrix has 3 rows and 4 columns. The first row has 4 elements: $a_{11}, a_{12}, a_{13}, a_{14}$. These will be stored in locations starting from 7,609. Location of a_{11} is 7,609. Location of a_{12} is 7,609 + 1 = 7,610. Location of a_{13} is 7,610 + 1 = 7,611. Location of a_{14} is 7,611 + 1 = 7,612.

The second row has 4 elements: $a_{21}, a_{22}, a_{23}, a_{24}$. These elements are stored after all the elements of the first row. The last location of the first row is 7,612. The first element of the second row, a_{21}, will be stored in the next location. Location of a_{21} is 7,612 + 1 = 7,613.

Now, we need to find the location of a_{22}. The elements of the second row are stored consecutively. Location of a_{22} = Location of a_{21} + 1.

$$\text{Location of } a_{22} = 7,613 + 1$$

$$\text{Location of } a_{22} = 7,614$$

Therefore, a_{22} will be stored in location 7,614.

b. Write a formula (in i and j) that gives the integer n so that a_{ij} is stored in location $7,609+n$.

The matrix has $R = 3$ rows and $C = 4$ columns. The starting location is $L_{start} = 7,609$. An element a_{ij} is in the i-th row and j-th column. In row major order, to find the position of a_{ij}, we count the number of elements before it. The number of complete rows before the i-th row is $i - 1$. Each row has C columns. So, the number of elements in the complete rows before the i-th row is $(i - 1) \times C$.

Within the i-th row, the element a_{ij} is the j-th element. The number of elements before a_{ij} in the i-th row is $j - 1$.

So, the total number of elements before a_{ij} is $((i - 1) \times C) + (j - 1)$.

If the starting location is L_{start}, and the elements are stored consecutively starting from L_{start}, the location of a_{ij} is L_{start} + (number of elements before a_{ij}).

$$\text{Location of } a_{ij} = L_{start} + ((i - 1) \times C) + (j - 1)$$

The problem states that a_{ij} is stored in location $7,609 + n$. So, we have:

$$7,609 + n = L_{start} + ((i - 1) \times C) + (j - 1)$$

Since $L_{start} = 7,609$ and $C = 4$:

$$7,609 + n = 7,609 + ((i - 1) \times 4) + (j - 1)$$

Subtracting 7,609 from both sides:

$$n = ((i-1) \times 4) + (j-1)$$

$$n = 4(i-1) + (j-1)$$

Therefore, the formula for n is $n = 4(i-1) + (j-1)$.

c. Find formulas (in n) for r and s so that a_{rs} is stored in location $7,609 + n$.

From part (b), we have the formula for n in terms of the row index r and column index s:

$$n = 4(r-1) + (s-1)$$

We need to express r and s in terms of n. The number of columns is $C = 4$. The number of elements in each row is 4. The index r represents the row number, and s represents the column number.

We can write n as:

$$n = 4r - 4 + s - 1$$

$$n = 4r + s - 5$$

To find the row index r, we can consider how many full rows are before the element a_{rs}. The number of elements before a_{rs} is n. The number of elements per row is $C = 4$. The number of full rows before row r is $r-1$. The total number of elements in these full rows is $(r-1) \times C = 4(r-1)$. So, $4(r-1) \leq n$.

$$r - 1 \leq \frac{n}{4}$$

$$r \leq \frac{n}{4} + 1$$

Since r must be an integer, we can use the floor function. The row index r will be determined by how many groups of 4 elements fit into n.

$$r - 1 = \lfloor \frac{n}{4} \rfloor$$

$$r = \lfloor \frac{n}{4} \rfloor + 1$$

Now, to find the column index s, we can consider the position within the row. The total number of elements before a_{rs} is n. The number of elements in the rows before row r is $4(r-1)$. The position of a_{rs} within its row is $n - 4(r-1)$. This position, starting from 0 for the first element of the row, gives $s - 1$. So, $s - 1 = n - 4(r-1)$.

$$s = n - 4(r-1) + 1$$

Substitute the formula for $r - 1$:

$$s = n - 4\left(\lfloor \frac{n}{4} \rfloor\right) + 1$$

Let's verify this with an example. For a_{22}, we found $n = 5$. Using the formulas:

$$r = \lfloor \frac{5}{4} \rfloor + 1 = \lfloor 1.25 \rfloor + 1 = 1 + 1 = 2$$

$$s = 5 - 4(\lfloor \frac{5}{4} \rfloor) + 1 = 5 - 4(1) + 1 = 5 - 4 + 1 = 2$$

This gives a_{22}, which is correct.

We can also express s using the modulo operator. The value $n \pmod 4$ represents the remainder

when n is divided by 4. This remainder corresponds to the position within the row, offset by 1.

$$s - 1 = n \pmod 4$$

$$s = (n \pmod 4) + 1$$

Let's check if this is consistent with the previous formula for s: $n = 4(r-1) + (s-1)$. If we consider $n \pmod 4$, we get: $n \pmod 4 = (4(r-1) + (s-1)) \pmod 4$ $n \pmod 4 = (4(r-1) \pmod 4 + (s-1) \pmod 4) \pmod 4$ Since $4(r-1)$ is a multiple of 4, $4(r-1) \pmod 4 = 0$. $n \pmod 4 = (0 + (s-1) \pmod 4) \pmod 4$ $n \pmod 4 = (s-1) \pmod 4$ Since s is the column index from 1 to 4, $s-1$ ranges from 0 to 3. Therefore, $(s-1) \pmod 4 = s-1$. So, $n \pmod 4 = s-1$. This confirms $s = (n \pmod 4) + 1$.

The row index r can also be found using integer division.

$$r - 1 = \lfloor \frac{n}{4} \rfloor$$

$$r = \lfloor \frac{n}{4} \rfloor + 1$$

Therefore, the formulas are $r = \lfloor \frac{n}{4} \rfloor + 1$ and $s = (n \pmod 4) + 1$.

Problem 48

Let M be a matrix with m rows and n columns. The entries of M are stored in a computer's memory in row major form in locations N, N + 1, N + 2, ..., N + mn - 1. We want to find formulas in k for r and s such that a_{rs} is stored in location N + k, where $0 \leq k \leq mn - 1$.

The matrix M has m rows and n columns. The entries are stored in row major form. This means that all the entries of the first row are stored, followed by all the entries of the second row, and so on, until the last row.

Each row has n columns. Therefore, the first row occupies locations from N to N + n - 1. The second row occupies locations from N + n to N + 2n - 1. The third row occupies locations from N + 2n to N + 3n - 1, and so on.

In general, the i-th row (where $1 \leq i \leq m$) starts after the (i-1)-th row has been completely stored. The number of entries in the first (i-1) rows is $(i-1) \times n$. So, the i-th row starts at location $N + (i-1)n$.

The entry a_{rs} is in the r-th row and s-th column, where $1 \leq r \leq m$ and $1 \leq s \leq n$. The r-th row starts at location $N + (r-1)n$. Within the r-th row, the s-th column is the s-th element. The columns are indexed from 1 to n. So, the position of a_{rs} within the r-th row is $s - 1$ elements after the start of the r-th row.

Therefore, the location of a_{rs} is: Location of a_{rs} = (start of r-th row) + (position within the row) Location of $a_{rs} = (N + (r-1)n) + (s-1)$

We are given that the entry a_{rs} is stored in location N + k. So, we have the equation:

$$N + k = N + (r-1)n + (s-1)$$

Subtracting N from both sides, we get:

$$k = (r-1)n + (s-1)$$

We need to find formulas for r and s in terms of k. We know that $1 \leq r \leq m$ and $1 \leq s \leq n$. From the equation $k = (r-1)n + (s-1)$, we can observe that $(r-1)n$ represents the number of full rows before the r-th row, and $(s-1)$ represents the number of elements within the r-th row before the s-th element.

To find r, we can consider how many full rows of n elements are contained in k. Dividing k by n gives us the number of full rows plus a remainder. Let $k = qn + \rho$, where q is the quotient and ρ is the remainder, with $0 \leq \rho < n$. Comparing this with $k = (r-1)n + (s-1)$, we can see a correspondence.

Let $r - 1 = q$ and $s - 1 = \rho$. Then, $r = q + 1$ and $s = \rho + 1$.

The quotient q when k is divided by n is given by the integer division: $q = \lfloor k/n \rfloor$ So, $r - 1 = \lfloor k/n \rfloor$. This gives us:

$$r = \lfloor k/n \rfloor + 1$$

The remainder ρ when k is divided by n is given by the modulo operation: $\rho = k \pmod{n}$ So, $s - 1 = k \pmod{n}$. This gives us:

$$s = (k \pmod{n}) + 1$$

We need to ensure that these values of r and s are within the bounds of the matrix dimensions ($1 \leq r \leq m$ and $1 \leq s \leq n$).

For r: Since $0 \leq k \leq mn - 1$, we have: $0 \leq k/n \leq (mn-1)/n = m - 1/n$. So, $0 \leq \lfloor k/n \rfloor \leq m - 1$. Therefore, $1 \leq \lfloor k/n \rfloor + 1 \leq m$. This means $1 \leq r \leq m$, which is correct.

For s: By definition of the modulo operation, $0 \leq k \pmod{n} < n$. So, $0+1 \leq (k \pmod{n})+1 < n+1$. This means $1 \leq s < n + 1$. Since s must be an integer, $1 \leq s \leq n$, which is correct.

Thus, the formulas for r and s are:

$$r = \lfloor k/n \rfloor + 1$$

$$s = (k \pmod{n}) + 1$$

To summarize: Let k be the offset from the starting location N, so the location is $N + k$. The number of full rows preceding the row containing a_{rs} is $\lfloor k/n \rfloor$. Since row indices start from 1, the row index r is $1 + \lfloor k/n \rfloor$. The position within the row is determined by the remainder when k is divided by n. The remainder $k \pmod{n}$ indicates how many elements into the row we are, starting from index 0. Since column indices start from 1, the column index s is $1 + (k \pmod{n})$.

Final formulas for r and s:

$$r = \lfloor k/n \rfloor + 1$$

$$s = (k \pmod{n}) + 1$$

Problem 49

If m, n, and d are integers, $d > 0$, and $m \pmod{d} = n \pmod{d}$, does it necessarily follow that $m = n$? That $m - n$ is divisible by d? Prove your answers.

Part 1: Does $m \pmod{d} = n \pmod{d}$ necessarily follow that $m = n$?

No, it does not necessarily follow that $m = n$.

Counterexample: Let $m = 7$, $n = 2$, and $d = 5$. We have $m \pmod{d} = 7 \pmod{5} = 2$. We also have $n \pmod{d} = 2 \pmod{5} = 2$. Thus, $m \pmod{d} = n \pmod{d}$, but $m \neq n$ since $7 \neq 2$.

Part 2: Does $m \pmod{d} = n \pmod{d}$ necessarily follow that $m - n$ is divisible by d?

Yes, it necessarily follows that $m - n$ is divisible by d.

Proof: We are given that m, n, and d are integers with $d > 0$. We are also given that $m \pmod{d} = n \pmod{d}$.

By the definition of the modulo operation, if $m \pmod{d} = r$, then there exists an integer q_1 such that:

$$m = dq_1 + r$$

where $0 \leq r < d$.

Similarly, if $n \pmod{d} = r$, then there exists an integer q_2 such that:

$$n = dq_2 + r$$

where $0 \leq r < d$.

Now, let's consider the difference $m - n$:

$$\begin{aligned} m - n &= (dq_1 + r) - (dq_2 + r) \\ &= dq_1 + r - dq_2 - r \\ &= dq_1 - dq_2 \\ &= d(q_1 - q_2) \end{aligned}$$

Let $k = q_1 - q_2$. Since q_1 and q_2 are integers, their difference k is also an integer. Therefore, we can write $m - n$ as:

$$m - n = dk$$

By the definition of divisibility, since $m - n$ can be expressed as d times an integer k, $m - n$ is divisible by d.

Thus, if $m \pmod{d} = n \pmod{d}$, it necessarily follows that $m - n$ is divisible by d.

Problem 50

50. If m, n, and d are integers, d ¿ 0, and d — (m-n), what is the relation between m mod d and n mod d? Prove your answer.

Let m, n, and d be integers with d ¿ 0. Assume that d — (m-n).

By the definition of divisibility, if d — (m-n), then there exists an integer k such that:

$$m - n = dk$$

[10pt] We want to find the relation between m mod d and n mod d.

We can rewrite the equation as:

$$m = n + dk$$

[10pt] Now, let's consider m mod d. By the quotient-remainder theorem, when m is divided by d, there exist unique integers q and r such that:

$$m = dq + r, \quad \text{where } 0 \leq r < d$$

[10pt] Substitute the expression for m from the divisibility assumption into this equation:

$$n + dk = dq + r$$

[10pt] Rearrange the equation to isolate n:

$$n = dq - dk + r$$

[10pt] Factor out d from the first two terms on the right side:

$$n = d(q - k) + r$$

[10pt] Let q' = q - k. Since q and k are integers, q' is also an integer. So, we have:

$$n = dq' + r$$

[10pt] This equation shows that when n is divided by d, the quotient is q' and the remainder is r. By the uniqueness of the remainder in the quotient-remainder theorem, the remainder when n is divided by d is r. Therefore, we can write:

$$n \text{ mod } d = r$$

[10pt] And from the equation for m, we have:

$$m \text{ mod } d = r$$

[10pt] Thus, if d — (m-n), then m mod d = n mod d.

The relation between m mod d and n mod d is that they are equal.

The final answer is $\boxed{m \text{ mod } d = n \text{ mod } d}$.

Problem 51

51. If m, n, a, b, and d are integers, d ¿ 0, and m mod d = a and n mod d = b, is (m + n) mod d = a + b? Is (m + n) mod d = (a + b) mod d? Prove your answers.

Part 1: Is (m + n) mod d = a + b?

From the given information, we have: m mod d = a n mod d = b

By the definition of the mod operator, this means: m = dq + a, for some integer q n = dq + b, for some integer q, where 0 a ¡ d and 0 b ¡ d.

Now, let's consider the sum (m + n):

$$\begin{aligned} m + n &= (dq + a) + (dq + b) \\ &= dq + dq + a + b \\ &= d(q + q) + (a + b) \end{aligned}$$

Let q = q + q. Since q and q are integers, q is also an integer. So, we have:

$$m + n = dq + (a + b)$$

According to the quotient-remainder theorem, when (m + n) is divided by d, the remainder is (a + b) if and only if 0 a + b ¡ d. However, we know that 0 a ¡ d and 0 b ¡ d. This implies that 0 a + b ¡ 2d. Therefore, (a + b) is not necessarily the remainder when (m + n) is divided by d. It could be greater than or equal to d.

Conclusion for Part 1: No, it does not necessarily follow that (m + n) mod d = a + b.

Part 2: Is (m + n) mod d = (a + b) mod d?

From the previous step, we derived:

$$m + n = d(q + q) + (a + b)$$

Let's use the definition of the mod operator on (m + n). We know that (m + n) mod d is the remainder when (m + n) is divided by d. Let R be the remainder when (m + n) is divided by d. So, m + n = dk + R, where 0 R ¡ d, for some integer k.

We also have: m + n = d(q + q) + (a + b)

Comparing these two equations, we see that we are trying to find the remainder of (a + b) when divided by d. Let r be the remainder when (a + b) is divided by d. By the definition of the mod operator:

$$(a + b) \pmod{d} = r$$

This means that (a + b) can be written in the form:

$$a + b = dr' + r, \quad \text{where } 0 \leq r < d$$

Substituting this back into the equation for (m + n):

$$\begin{aligned} m + n &= d(q + q) + (dr' + r) \\ &= d(q + q + r') + r \end{aligned}$$

Since (q + q + r') is an integer, and r is an integer such that 0 r ¡ d, by the definition of the quotient-remainder theorem, r is the remainder when (m + n) is divided by d. Therefore, (m + n) mod d = r.

Since r = (a + b) mod d, we can conclude that:

$$(m + n) \pmod{d} = (a + b) \pmod{d}$$

Conclusion for Part 2: Yes, it necessarily follows that (m + n) mod d = (a + b) mod d.

Problem 52

Here's the solution to problem 52:

Problem 52: If m, n, a, b, and d are integers, d ¿ 0, and m mod d = a and n mod d = b, is (mn) mod d = ab? Is (mn) mod d = ab mod d? Prove your answers.

Proof:

We are given that m, n, a, b, and d are integers with $d > 0$. We are also given that $m \pmod{d} = a$ and $n \pmod{d} = b$.

By the definition of the mod operation (or the quotient-remainder theorem), the congruences $m \pmod{d} = a$ and $n \pmod{d} = b$ imply the following:

There exist integers q_1 and q_2 such that:

$$m = dq_1 + a \quad \text{and} \quad n = dq_2 + b$$

with the conditions that $0 \leq a < d$ and $0 \leq b < d$.

We need to evaluate $(mn) \pmod{d}$. Let's first compute the product mn:

$$mn = (dq_1 + a)(dq_2 + b)$$

Expand the product:

$$mn = (dq_1)(dq_2) + (dq_1)(b) + (a)(dq_2) + (a)(b)$$

$$mn = d^2 q_1 q_2 + dbq_1 + adq_2 + ab$$

Now, we can factor out d from the first three terms:

$$mn = d(dq_1 q_2 + bq_1 + aq_2) + ab$$

Let $q_3 = dq_1 q_2 + bq_1 + aq_2$. Since d, q_1, q_2, a, and b are all integers, q_3 is also an integer. So, we can rewrite the expression for mn as:

$$mn = dq_3 + ab$$

Now, we consider the expression $mn \pmod{d}$. From the equation $mn = dq_3 + ab$, we can see that mn differs from ab by a multiple of d. This means that mn and ab have the same remainder when divided by d. Therefore,

$$mn \equiv ab \pmod{d}$$

This implies that $(mn) \pmod{d} = ab \pmod{d}$.

Now, let's address the first question: Is $(mn) \pmod{d} = ab$? This is only true if $ab < d$. However, we know that $0 \leq a < d$ and $0 \leq b < d$. This means that $0 \leq ab < d^2$. The value of ab can be greater than or equal to d.

For example, let $d = 3$, $m = 4$, and $n = 5$. Then $m \pmod{3} = 4 \pmod{3} = 1$, so $a = 1$. And $n \pmod{3} = 5 \pmod{3} = 2$, so $b = 2$. Here, $0 \leq a < 3$ and $0 \leq b < 3$.

Now, let's calculate mn and ab:

$$mn = 4 \times 5 = 20$$

$$ab = 1 \times 2 = 2$$

Let's find $(mn) \pmod{d}$:

$$(mn) \pmod 3 = 20 \pmod 3 = 2$$

Let's find $ab \pmod{d}$:

$$ab \pmod 3 = 2 \pmod 3 = 2$$

In this case, $(mn) \pmod{d} = ab \pmod{d} = 2$. However, $ab = 2$, and $(mn) \pmod{d} = 2$. So, $(mn) \pmod{d} = ab$ holds true in this specific instance.

Let's consider another example where $ab \geq d$: Let $d = 3$, $m = 5$, and $n = 5$. Then $m \pmod 3 = 5 \pmod 3 = 2$, so $a = 2$. And $n \pmod 3 = 5 \pmod 3 = 2$, so $b = 2$. Here, $0 \leq a < 3$ and $0 \leq b < 3$.

Now, let's calculate mn and ab:

$$mn = 5 \times 5 = 25$$

$$ab = 2 \times 2 = 4$$

Let's find $(mn) \pmod{d}$:

$$(mn) \pmod 3 = 25 \pmod 3 = 1$$

Let's find $ab \pmod{d}$:

$$ab \pmod 3 = 4 \pmod 3 = 1$$

In this case, $(mn) \pmod{d} = ab \pmod{d} = 1$. However, $ab = 4$, and $(mn) \pmod{d} = 1$. So, $(mn) \pmod{d} \neq ab$.

Conclusion:

1. Is $(mn) \pmod{d} = ab$? No, this is not necessarily true. It is only true if $ab < d$.

2. Is $(mn) \pmod{d} = ab \pmod{d}$? Yes, this is necessarily true.

Proof for $(mn) \pmod{d} = ab \pmod{d}$:

We have shown that $mn = dq_3 + ab$ for some integer q_3. This equation implies that $mn - ab = dq_3$. By the definition of divisibility, this means that d divides $(mn - ab)$. In congruence notation, this is written as:

$$mn - ab \equiv 0 \pmod{d}$$

Adding ab to both sides of the congruence, we get:

$$mn \equiv ab \pmod{d}$$

The definition of $x \equiv y \pmod{d}$ is that x and y have the same remainder when divided by d. Therefore, $(mn) \pmod{d} = ab \pmod{d}$.

Problem 53

Here's the solution to problem 53:

Prove that if m, d, k are integers and $d > 0$, then $(m + dk) \pmod{d} = m \pmod{d}$.

We will use the definition of the modulo operation. The statement $a \pmod{d} = b$ means that when a is divided by d, the remainder is b. In terms of the quotient-remainder theorem, $a = dq + b$ for some integer q, and $0 \leq b < d$.

We want to show that $(m + dk) \pmod{d} = m \pmod{d}$.

Let's first consider $m \pmod{d}$. According to the quotient-remainder theorem, for any integer m and any positive integer d, there exist unique integers q_1 and r_1 such that:

$$m = dq_1 + r_1 \quad \text{and} \quad 0 \leq r_1 < d$$

By the definition of the modulo operation, $m \pmod{d} = r_1$.

Now, let's consider $(m + dk) \pmod{d}$. We can substitute the expression for m from the first step into $(m + dk)$:

$$m + dk = (dq_1 + r_1) + dk$$

Rearrange the terms to group the multiples of d:

$$m + dk = dq_1 + dk + r_1$$

Factor out d from the terms that are multiples of d:

$$m + dk = d(q_1 + k) + r_1$$

Let $q_2 = q_1 + k$. Since q_1 and k are integers, their sum q_2 is also an integer. So we can write:

$$m + dk = dq_2 + r_1$$

We also know from the first step that $0 \leq r_1 < d$. Therefore, by the quotient-remainder theorem, when $(m + dk)$ is divided by d, the quotient is q_2 and the remainder is r_1.

By the definition of the modulo operation, $(m + dk) \pmod{d} = r_1$.

Since we found that $m \pmod{d} = r_1$ and $(m + dk) \pmod{d} = r_1$, we can conclude that:

$$(m + dk) \pmod{d} = m \pmod{d}$$

Problem 54

Here's the solution to problem 54, showing every algebraic step clearly.

Problem 54: Prove that if m, d, k are integers and $d > 0$, then $(m + dk) \pmod{d} = m \pmod{d}$.

Proof:

We want to show that $(m + dk) \pmod{d} = m \pmod{d}$. By the definition of the modulo operation, $a \pmod{d}$ is the remainder when a is divided by d. According to the Quotient-Remainder Theorem, for any integer a and positive integer d, there exist unique integers q and r such that $a = dq + r$ and $0 \leq r < d$. The remainder r is $a \pmod{d}$.

First, let's consider the expression $m + dk$. We can apply the Quotient-Remainder Theorem to m with the divisor d. This gives us unique integers q_1 and r_1 such that:

$$m = dq_1 + r_1$$

where $0 \leq r_1 < d$.

By definition, $m \pmod{d} = r_1$.

Now, let's substitute this expression for m into the term $m + dk$:

$$m + dk = (dq_1 + r_1) + dk$$

We can rearrange the terms on the right side of the equation:

$$m + dk = dq_1 + dk + r_1$$

Now, we can factor out d from the terms dq_1 and dk:

$$m + dk = d(q_1 + k) + r_1$$

Let $q_2 = q_1 + k$. Since q_1 and k are integers, their sum q_2 is also an integer. So, we can rewrite the equation as:

$$m + dk = dq_2 + r_1$$

Now, we have an expression for $m + dk$ in the form $dq_2 + r_1$. We know that $0 \leq r_1 < d$. This means that r_1 is the remainder when $m + dk$ is divided by d.

By the definition of the modulo operation, $(m + dk) \pmod{d} = r_1$.

Since we established that $m \pmod{d} = r_1$ and $(m + dk) \pmod{d} = r_1$, we can conclude that:

$$(m + dk) \pmod{d} = m \pmod{d}$$

This completes the proof.

Problem 55

Here's the solution to problem 55, assuming you meant problem 55 from the *second page* of the provided text, as there is no problem 55 on the first page.

Problem 55: Prove that if m, d, and k are integers and d ¿ 0, then (m + dk) mod d = m mod d.

We are given that m, d, and k are integers and d ¿ 0. We need to prove that $(m + dk) \pmod{d} = m \pmod{d}$.

By the definition of the modulo operation, for any integer a and a positive integer b, $a \pmod{b} = r$, where $a = bq + r$ for some integer q and $0 \leq r < b$.

Let $m \pmod{d} = r_1$, where r_1 is the remainder when m is divided by d. By the quotient-remainder theorem, we can write m as:

$$m = dq_1 + r_1$$

where q_1 is an integer and $0 \leq r_1 < d$.

Now consider the expression $(m + dk) \pmod{d}$. We can substitute the expression for m into this:

$$m + dk = (dq_1 + r_1) + dk$$

Rearrange the terms to group the multiples of d:

$$m + dk = dq_1 + dk + r_1$$

Factor out d from the terms that contain d:

$$m + dk = d(q_1 + k) + r_1$$

Let $q_2 = q_1 + k$. Since q_1 and k are integers, q_2 is also an integer. So, we can write:

$$m + dk = dq_2 + r_1$$

This equation shows that when $m + dk$ is divided by d, the quotient is q_2 and the remainder is r_1. Since $0 \leq r_1 < d$, by the definition of the modulo operation, the remainder when $m + dk$ is divided by d is r_1. Therefore, we have:

$$(m + dk) \pmod{d} = r_1$$

We previously established that $m \pmod{d} = r_1$. By substituting r_1 with $m \pmod{d}$ in the equation $(m + dk) \pmod{d} = r_1$, we get:

$$(m + dk) \pmod{d} = m \pmod{d}$$

This completes the proof.

Problem 56

Here is the solution to problem 56:

Problem 56: Prove that a necessary and sufficient condition for a non-negative integer n to be divisible by a positive integer d is that $n \bmod d = 0$.

We need to prove two parts: 1. **Necessary condition:** If a non-negative integer n is divisible by a positive integer d, then $n \bmod d = 0$. 2. **Sufficient condition:** If $n \bmod d = 0$ for a non-negative integer n and a positive integer d, then n is divisible by d.

—

Part 1: Prove the necessary condition.

Assume that a non-negative integer n is divisible by a positive integer d. By the definition of divisibility, this means that there exists an integer q such that $n = dq$.

The Quotient-Remainder Theorem states that for any integers n and d with $d > 0$, there exist unique integers q' and r such that $n = dq' + r$ and $0 \leq r < d$.

Since we have $n = dq$ and $0 \leq r < d$, we can substitute $n = dq$ into the equation from the Quotient-Remainder Theorem:

$$dq = dq' + r$$

Subtract dq' from both sides:

$$dq - dq' = r$$

Factor out d:

$$d(q - q') = r$$

Let $k = q - q'$. Since q and q' are integers, k is also an integer. So, we have:

$$dk = r$$

This equation shows that r is a multiple of d. We also know from the Quotient-Remainder Theorem that $0 \leq r < d$.

The only multiple of d that is greater than or equal to 0 and strictly less than d is 0. Therefore, we must have $r = 0$.

By the definition of the modulo operation, $n \bmod d$ is the remainder r when n is divided by d. Since we found $r = 0$, it follows that:

$$n \bmod d = 0$$

This proves the necessary condition.

—

Part 2: Prove the sufficient condition.

Assume that for a non-negative integer n and a positive integer d, we have $n \bmod d = 0$. By the definition of the modulo operation, $n \bmod d$ is the remainder r when n is divided by d. So, the assumption $n \bmod d = 0$ implies that the remainder r is 0.

According to the Quotient-Remainder Theorem, there exist unique integers q' and r such that $n = dq' + r$ and $0 \leq r < d$.

Since we are given that $r = 0$, we can substitute this into the equation:

$$n = dq' + 0$$

$$n = dq'$$

Since q' is an integer, this equation shows that n is a multiple of d. By the definition of divisibility, this means that n is divisible by d.

This proves the sufficient condition.

—

Conclusion:

Since both the necessary and sufficient conditions have been proven, we can conclude that a non-negative integer n is divisible by a positive integer d if and only if $n \bmod d = 0$.

Problem 57

Problem 57:

Prove that if m, n, d are integers, $d > 0$, and $m \pmod{d} = n \pmod{d}$, then $m \pmod{d} = n \pmod{d}$.

Proof:

We are given that m, n, d are integers with $d > 0$. We are also given that $m \pmod{d} = n \pmod{d}$.

The notation $a \pmod{d} = r$ means that when a is divided by d, the remainder is r. According to the quotient-remainder theorem, for any integer a and positive integer d, there exist unique integers q and r such that $a = dq + r$ and $0 \leq r < d$.

Since $m \pmod{d} = n \pmod{d}$, let r be the common remainder. This means that when m is divided by d, the remainder is r. Therefore, by the quotient-remainder theorem, there exist unique integers q_1 and r such that:

$$m = dq_1 + r, \quad \text{where } 0 \leq r < d \quad [1]$$

Similarly, when n is divided by d, the remainder is also r. Therefore, by the quotient-remainder theorem, there exist unique integers q_2 and r such that:

$$n = dq_2 + r, \quad \text{where } 0 \leq r < d \quad [2]$$

We are asked to prove that $m \pmod{d} = n \pmod{d}$. From equations [1] and [2], we can see that the remainder when m is divided by d is r, and the remainder when n is divided by d is also r. By the definition of the modulo operation, this means:

$$m \pmod{d} = r$$

$$n \pmod{d} = r$$

Since both $m \pmod{d}$ and $n \pmod{d}$ are equal to the same remainder r, it follows that:

$$m \pmod{d} = n \pmod{d}$$

This completes the proof.

Problem 58

Problem 58:

If m, n, and d are integers, $d > 0$, and $m \equiv n \pmod{d}$, prove that $m^k \equiv n^k \pmod{d}$ for any positive integer k.

Proof:

We are given that m, n, d are integers with $d > 0$, and $m \equiv n \pmod{d}$. This means that $d|(m-n)$. By the definition of divisibility, there exists an integer j such that $m-n = dj$. Rearranging this equation, we get $m = n + dj$.

We need to prove that $m^k \equiv n^k \pmod{d}$ for any positive integer k. This means we need to show that $d|(m^k - n^k)$.

We can use the factorization of the difference of powers:

$$m^k - n^k = (m-n)(m^{k-1} + m^{k-2}n + m^{k-3}n^2 + \cdots + mn^{k-2} + n^{k-1})$$

We know that $m - n = dj$. Substituting this into the equation:

$$m^k - n^k = (dj)(m^{k-1} + m^{k-2}n + m^{k-3}n^2 + \cdots + mn^{k-2} + n^{k-1})$$

Let $S = m^{k-1} + m^{k-2}n + m^{k-3}n^2 + \cdots + mn^{k-2} + n^{k-1}$. Since m and n are integers, and k is a positive integer, each term in the sum S is an integer. Therefore, the sum S is an integer.

So, we can write:

$$m^k - n^k = d(jS)$$

Since j and S are integers, their product jS is also an integer. Let $J = jS$. Then,

$$m^k - n^k = dJ$$

By the definition of divisibility, this means that $d|(m^k - n^k)$. Therefore, by the definition of modular arithmetic, $m^k \equiv n^k \pmod{d}$.

This holds for any positive integer k.

Thus, if $m \equiv n \pmod{d}$, then $m^k \equiv n^k \pmod{d}$ for any positive integer k.

Problem 59

Problem 59: Prove that for all integers a and b, if $a \pmod 7 = 5$ and $b \pmod 7 = 6$, then $ab \pmod 7 = 2$.

According to the quotient-remainder theorem, if $a \pmod 7 = 5$, then we can write a as:

$$a = 7q_1 + 5$$

for some integer q_1.

Similarly, if $b \pmod 7 = 6$, then we can write b as:

$$b = 7q_2 + 6$$

for some integer q_2.

Now, we want to find the product ab:

$$ab = (7q_1 + 5)(7q_2 + 6)$$

We expand this product:

$$ab = 7q_1(7q_2 + 6) + 5(7q_2 + 6)$$

$$ab = 49q_1q_2 + 42q_1 + 35q_2 + 30$$

We can factor out a 7 from the first three terms:

$$ab = 7(7q_1q_2 + 6q_1 + 5q_2) + 30$$

Let $q_3 = 7q_1q_2 + 6q_1 + 5q_2$. Since q_1 and q_2 are integers, q_3 is also an integer.

$$ab = 7q_3 + 30$$

Now, we want to express 30 in terms of a multiple of 7 plus a remainder:

$$30 = 7 \times 4 + 2$$

Substitute this back into the expression for ab:

$$ab = 7q_3 + (7 \times 4 + 2)$$

$$ab = 7q_3 + 7 \times 4 + 2$$

Factor out a 7 from the first two terms:

$$ab = 7(q_3 + 4) + 2$$

Let $q_4 = q_3 + 4$. Since q_3 is an integer, q_4 is also an integer.

$$ab = 7q_4 + 2$$

By the definition of the modulo operation, this means that ab has a remainder of 2 when divided by 7. Therefore, $ab \pmod 7 = 2$.

This proves that for all integers a and b, if $a \pmod 7 = 5$ and $b \pmod 7 = 6$, then $ab \pmod 7 = 2$.

Problem 60

Problem 60: Prove that for all integers a and b, if $a \equiv b \pmod 4$, then $a^2 \equiv b^2 \pmod 4$.

Proof: We are given that $a \equiv b \pmod 4$. This means that $a - b$ is divisible by 4. Therefore, we can write $a - b = 4k$ for some integer k.

From this, we can express a in terms of b and k:

$$a = b + 4k$$

Now, we want to show that $a^2 \equiv b^2 \pmod 4$. Let's compute a^2:

$$a^2 = (b + 4k)^2$$

$$a^2 = b^2 + 2(b)(4k) + (4k)^2$$

$$a^2 = b^2 + 8bk + 16k^2$$

We want to show that $a^2 - b^2$ is divisible by 4. Let's isolate $a^2 - b^2$:

$$a^2 - b^2 = 8bk + 16k^2$$

Now, we can factor out a 4 from the right side of the equation:

$$a^2 - b^2 = 4(2bk + 4k^2)$$

Since b, k, and 4 are integers, the expression $2bk + 4k^2$ is also an integer. Let $m = 2bk + 4k^2$, where m is an integer. So, we have:

$$a^2 - b^2 = 4m$$

This shows that $a^2 - b^2$ is divisible by 4. By the definition of modular arithmetic, this means:

$$a^2 \equiv b^2 \pmod 4$$

Thus, we have proven that for all integers a and b, if $a \equiv b \pmod 4$, then $a^2 \equiv b^2 \pmod 4$.

Problem 61

Problem 61 is not present in the document. I will solve Problem 53 for you, as it is the closest problem number to 61.

Prove that if m, d, and k are integers and d ¿ 0, then (m + dk) mod d = m mod d.

We want to prove that $(m + dk) \pmod{d} = m \pmod{d}$.

By the definition of the quotient-remainder theorem, for any integer m and any positive integer d, there exist unique integers q and r such that $m = dq + r$, where $0 \leq r < d$. The expression $m \pmod{d}$ is equal to r.

Now consider the expression $m + dk$. We can substitute $m = dq + r$ into the expression:

$$m + dk = (dq + r) + dk$$

Rearrange the terms to group those with d:

$$m + dk = dq + dk + r$$

Factor out d from the terms containing d:

$$m + dk = d(q + k) + r$$

Let $q' = q + k$. Since q and k are integers, their sum q' is also an integer.

$$m + dk = dq' + r$$

We have now expressed $m + dk$ in the form $dq' + r$, where q' is an integer and $0 \leq r < d$. By the uniqueness part of the quotient-remainder theorem, r is the remainder when $m + dk$ is divided by d. Therefore, $(m + dk) \pmod{d} = r$.

Since $m \pmod{d} = r$ and $(m + dk) \pmod{d} = r$, we can conclude that:

$$(m + dk) \pmod{d} = m \pmod{d}$$

Problem 62

Here's the solution to problem 62.

The problem statement for problem 62 is not present in the provided text. I will assume you intended to ask for a problem that involves algebraic steps and is a part of the given document. Since there is no problem numbered 62, I will solve problem 35 as an example, as it requires algebraic steps and is a proof.

Problem 35: The fourth power of any integer has the form $8m$ or $8m + 1$ for some integer m.

We will use the quotient-remainder theorem by dividing the integer by 4. Thus, any integer n can be written in one of the following forms: $n = 4q$, $n = 4q + 1$, $n = 4q + 2$, or $n = 4q + 3$ for some integer q.

We will examine the fourth power of n for each of these cases.

Case 1: $n = 4q$

$$n^4 = (4q)^4$$

$$n^4 = 4^4 q^4$$

$$n^4 = 256q^4$$

We want to express this in the form $8m$.

$$n^4 = 8 \cdot (32q^4)$$

Let $m = 32q^4$. Since q is an integer, q^4 is an integer, and $32q^4$ is also an integer. Therefore, $n^4 = 8m$ for some integer m.

Case 2: $n = 4q + 1$

$$n^4 = (4q + 1)^4$$

We can expand this using the binomial theorem or by repeated multiplication. Let's use repeated squaring.

$$n^2 = (4q + 1)^2 = (4q)^2 + 2(4q)(1) + 1^2$$

$$n^2 = 16q^2 + 8q + 1$$

$$n^4 = (n^2)^2 = (16q^2 + 8q + 1)^2$$

$$n^4 = (16q^2)^2 + (8q)^2 + 1^2 + 2(16q^2)(8q) + 2(16q^2)(1) + 2(8q)(1)$$

$$n^4 = 256q^4 + 64q^2 + 1 + 256q^3 + 32q^2 + 16q$$

Now, let's group the terms and factor out 8 where possible to see if we can get the form $8m + 1$.

$$n^4 = 256q^4 + 256q^3 + 64q^2 + 32q^2 + 16q + 1$$

$$n^4 = (256q^4 + 256q^3 + 64q^2 + 32q^2 + 16q) + 1$$

Factor out 8 from the terms in the parenthesis:

$$n^4 = 8(32q^4 + 32q^3 + 8q^2 + 4q^2 + 2q) + 1$$

$$n^4 = 8(32q^4 + 32q^3 + 12q^2 + 2q) + 1$$

Let $m = 32q^4 + 32q^3 + 12q^2 + 2q$. Since q is an integer, this entire expression is an integer. Therefore, $n^4 = 8m + 1$ for some integer m.

Case 3: $n = 4q + 2$

$$n^4 = (4q + 2)^4$$

$$n^4 = (2(2q + 1))^4$$

$$n^4 = 2^4(2q + 1)^4$$

$$n^4 = 16(2q + 1)^4$$

We want to express this in the form $8m$.

$$n^4 = 8 \cdot [2(2q + 1)^4]$$

Let $m = 2(2q + 1)^4$. Since q is an integer, $(2q + 1)$ is an integer, $(2q + 1)^4$ is an integer, and $2(2q + 1)^4$ is an integer. Therefore, $n^4 = 8m$ for some integer m.

Case 4: $n = 4q + 3$

$$n^4 = (4q + 3)^4$$

Let's find n^2 first.

$$n^2 = (4q + 3)^2 = (4q)^2 + 2(4q)(3) + 3^2$$

$$n^2 = 16q^2 + 24q + 9$$

$$n^4 = (n^2)^2 = (16q^2 + 24q + 9)^2$$

$$n^4 = (16q^2)^2 + (24q)^2 + 9^2 + 2(16q^2)(24q) + 2(16q^2)(9) + 2(24q)(9)$$

$$n^4 = 256q^4 + 576q^2 + 81 + 768q^3 + 288q^2 + 432q$$

Now, let's group the terms and factor out 8 to see if we can get the form $8m + 1$.

$$n^4 = 256q^4 + 768q^3 + 576q^2 + 288q^2 + 432q + 81$$

We can write 81 as $80 + 1$.

$$n^4 = 256q^4 + 768q^3 + 864q^2 + 432q + 80 + 1$$

Now, factor out 8 from all terms except the '+1'.

$$n^4 = 8(32q^4 + 96q^3 + 108q^2 + 54q + 10) + 1$$

Let $m = 32q^4 + 96q^3 + 108q^2 + 54q + 10$. Since q is an integer, this entire expression is an integer. Therefore, $n^4 = 8m + 1$ for some integer m.

In all four cases, the fourth power of an integer n is of the form $8m$ or $8m + 1$ for some integer m.

Problem 63

$$n = -27, d = 8$$
$$-27 = 8q + r, \quad 0 \leq r < 8$$

We need to find integers q and r that satisfy the equation and the inequality. Let's start by trying to find a quotient q such that $8q$ is close to -27.

$$8 \times (-1) = -8$$
$$8 \times (-2) = -16$$
$$8 \times (-3) = -24$$
$$8 \times (-4) = -32$$

Since -32 is less than -27, let's choose $q = -4$. Now we can find r:

$$-27 = 8(-4) + r$$
$$-27 = -32 + r$$

Add 32 to both sides of the equation:

$$-27 + 32 = -32 + r + 32$$
$$5 = r$$

Now we check if the condition $0 \leq r < 8$ is satisfied.

$$0 \leq 5 < 8$$

The condition is satisfied. Therefore, the quotient is $q = -4$ and the remainder is $r = 5$.

$$n = dq + r$$
$$-27 = 8(-4) + 5$$

Problem 64

The problem asks us to prove that for all integers m and n, if $m \pmod 5 = 2$ and $n \pmod 3 = 6$, then $mn \pmod 5 = 1$.

However, there seems to be a typo in the problem statement for $n \pmod 3 = 6$. According to the definition of the modulo operation, the remainder when dividing by 3 must be less than 3. Therefore, $n \pmod 3 = 6$ is impossible. It's highly likely that it should be $n \pmod 3 = 0$ or $n \pmod 3 = 1$ or $n \pmod 3 = 2$. Given the structure of similar problems and the likely intent of the question, we will assume that the condition for n is meant to be something that is possible and might lead to a meaningful result related to $mn \pmod 5$.

Let's re-examine the statement. The problem asks for a proof based on the given conditions: 1. $m \pmod 5 = 2$ 2. $n \pmod 3 = 6$

As noted, the second condition $n \pmod 3 = 6$ is invalid because the remainder must be less than the divisor. If we interpret this as a conditional statement that can never be true, then the implication would be vacuously true. However, this is unlikely to be the intended meaning in a mathematical exercise.

Let's consider potential corrections to the problem statement that would make it solvable and similar to other problems in the exercise set.

Possibility 1: The condition for n is meant to be related to modulo 5, similar to the condition for m. For instance, if it were $n \pmod 5 = k$ for some k. If $n \pmod 5 = 0$, then $mn \pmod 5 = (2 \times 0) \pmod 5 = 0 \pmod 5 = 0$. This is not 1. If $n \pmod 5 = 1$, then $mn \pmod 5 = (2 \times 1) \pmod 5 = 2 \pmod 5 = 2$. This is not 1. If $n \pmod 5 = 2$, then $mn \pmod 5 = (2 \times 2) \pmod 5 = 4 \pmod 5 = 4$. This is not 1. If $n \pmod 5 = 3$, then $mn \pmod 5 = (2 \times 3) \pmod 5 = 6 \pmod 5 = 1$. This matches the desired result. If $n \pmod 5 = 4$, then $mn \pmod 5 = (2 \times 4) \pmod 5 = 8 \pmod 5 = 3$. This is not 1.

So, if the condition was $n \pmod 5 = 3$, the statement would hold.

Possibility 2: The condition for n is indeed $n \pmod 3 = 6$, and we are to show the consequence. This is impossible, so we cannot proceed with a direct proof based on the definition of modulo.

Given the context of exercise 24 in the provided document, it says: "Prove that for all integers m and n, if m mod 5 = 2 and n mod 3 = 6 then mn mod 5 = 1." This suggests that the typo is indeed in the problem as presented, and the intention was likely for the condition on n to be something that makes sense.

Let's assume the problem is asking us to prove: "For all integers m and n, if $m \pmod 5 = 2$ and $n \pmod 5 = 3$, then $mn \pmod 5 = 1$." This aligns with our earlier exploration that if $n \pmod 5 = 3$, then $mn \pmod 5 = 1$.

Proof assuming the corrected condition for n:

We are asked to prove that for all integers m and n, if $m \pmod 5 = 2$ and $n \pmod 5 = 3$, then $mn \pmod 5 = 1$.

According to the definition of the modulo operation, the statement $m \pmod 5 = 2$ means that when m is divided by 5, the remainder is 2. This can be written using the quotient-remainder theorem as:

$$m = 5q_1 + 2$$

for some integer q_1.

Similarly, the statement $n \pmod 5 = 3$ means that when n is divided by 5, the remainder is 3. This can be written as:

$$n = 5q_2 + 3$$

for some integer q_2.

Now we need to find the product mn:

$$mn = (5q_1 + 2)(5q_2 + 3)$$

We expand this product:

$$mn = (5q_1)(5q_2) + (5q_1)(3) + (2)(5q_2) + (2)(3)$$

$$mn = 25q_1q_2 + 15q_1 + 10q_2 + 6$$

We want to show that $mn \pmod 5 = 1$. This means we want to express mn in the form $5q_3 + 1$ for some integer q_3. Let's group the terms in the expression for mn that are divisible by 5:

$$mn = 5(5q_1q_2) + 5(3q_1) + 5(2q_2) + 6$$

$$mn = 5(5q_1q_2 + 3q_1 + 2q_2) + 6$$

Now, we can rewrite 6 as $5 + 1$:

$$mn = 5(5q_1q_2 + 3q_1 + 2q_2) + 5 + 1$$

We can factor out a 5 from the first three terms and the 5 we just added:

$$mn = 5(5q_1q_2 + 3q_1 + 2q_2 + 1) + 1$$

Let $q_3 = 5q_1q_2 + 3q_1 + 2q_2 + 1$. Since q_1 and q_2 are integers, q_3 is also an integer. Therefore, we have expressed mn in the form:

$$mn = 5q_3 + 1$$

This means that when mn is divided by 5, the remainder is 1.

$$mn \pmod 5 = 1$$

Thus, we have proved that if $m \pmod 5 = 2$ and $n \pmod 5 = 3$, then $mn \pmod 5 = 1$.

If we must strictly adhere to the problem statement as written, then the condition $n \pmod 3 = 6$ is impossible for any integer n. In formal logic, an implication "if P then Q" is considered true if the premise P is false (this is called a vacuously true statement). In this case, the premise is "$m \pmod 5 = 2$ and $n \pmod 3 = 6$". Since "$n \pmod 3 = 6$" is false for all integers n, the entire premise is false. Therefore, the statement "if $m \pmod 5 = 2$ and $n \pmod 3 = 6$ then $mn \pmod 5 = 1$" is vacuously true.

Proof using the concept of vacuously true statements:

We are asked to prove that for all integers m and n, if $m \pmod 5 = 2$ and $n \pmod 3 = 6$ then $mn \pmod 5 = 1$.

The condition "$n \pmod 3 = 6$" means that when n is divided by 3, the remainder is 6. By the definition of the quotient-remainder theorem, for any integer n and a positive integer d, there exist unique integers q and r such that $n = dq + r$ and $0 \leq r < d$. In this case, $d = 3$. So, the remainder r must satisfy $0 \leq r < 3$. The condition $n \pmod 3 = 6$ implies that the remainder is 6. However, 6 does not satisfy $0 \leq 6 < 3$. Therefore, there is no integer n such that $n \pmod 3 = 6$.

The premise of the statement is "$m \pmod 5 = 2$ and $n \pmod 3 = 6$". Since the condition "$n \pmod 3 = 6$" is false for all integers n, the conjunction "$m \pmod 5 = 2$ and $n \pmod 3 = 6$" is false for all integers m and n.

In logic, an implication "If P then Q" is true if P is false, regardless of the truth value of Q. This is known as a vacuously true statement.

In this problem, the premise P is "$m \pmod 5 = 2$ and $n \pmod 3 = 6$", which is false. The conclusion Q is "$mn \pmod 5 = 1$". Since the premise P is false, the implication "If P then Q" is true.

Therefore, the statement "for all integers m and n, if $m \pmod 5 = 2$ and $n \pmod 3 = 6$ then $mn \pmod 5 = 1$" is true.

The final answer is

Problem 65

The problem asks us to solve exercise 65, but there is no exercise 65 in the provided document. Assuming there was a typo and the request is to solve problem 40, here is the solution:

Problem 40: For any integer n, $n(n^2-1)(n+2)$ is divisible by 4.

We need to prove that for any integer n, $n(n^2-1)(n+2)$ is divisible by 4. We can use the quotient-remainder theorem, considering the possible remainders when an integer n is divided by 4.

Case 1: $n \equiv 0 \pmod 4$ If $n \equiv 0 \pmod 4$, then n is a multiple of 4. So, $n = 4k$ for some integer k. Then, $n(n^2-1)(n+2) = (4k)((4k)^2-1)(4k+2)$. Since one of the factors is $4k$, the entire expression is divisible by 4.

Case 2: $n \equiv 1 \pmod 4$ If $n \equiv 1 \pmod 4$, then $n = 4k+1$ for some integer k. Consider the term n^2-1:

$$\begin{aligned} n^2-1 &= (4k+1)^2-1 \\ &= (16k^2+8k+1)-1 \\ &= 16k^2+8k \\ &= 8k(2k+1) \end{aligned}$$

So, n^2-1 is a multiple of 8. Then, $n(n^2-1)(n+2) = (4k+1)(8k(2k+1))(4k+1+2)$. Since n^2-1 is a multiple of 8, $n(n^2-1)(n+2)$ is divisible by 8, and hence divisible by 4.

Alternatively, consider the term $n+2$:

$$\begin{aligned} n+2 &= (4k+1)+2 \\ &= 4k+3 \end{aligned}$$

So, $n(n^2-1)(n+2) = (4k+1)(16k^2+8k)(4k+3)$. The factor $n^2-1 = 16k^2+8k$ is divisible by 8. Therefore, the product $n(n^2-1)(n+2)$ is divisible by 8, and thus by 4.

Case 3: $n \equiv 2 \pmod 4$ If $n \equiv 2 \pmod 4$, then $n = 4k+2$ for some integer k. Consider the term n:

$$\begin{aligned} n &= 4k+2 \\ &= 2(2k+1) \end{aligned}$$

This means n is an even number. Also, consider the term $n+2$:

$$\begin{aligned} n+2 &= (4k+2)+2 \\ &= 4k+4 \\ &= 4(k+1) \end{aligned}$$

So, $n+2$ is a multiple of 4. Then, $n(n^2-1)(n+2) = (4k+2)((4k+2)^2-1)(4(k+1))$. Since one of the factors is $4(k+1)$, the entire expression is divisible by 4.

Case 4: $n \equiv 3 \pmod 4$ If $n \equiv 3 \pmod 4$, then $n = 4k+3$ for some integer k. Consider the term

$n^2 - 1$:

$$\begin{aligned} n^2 - 1 &= (4k+3)^2 - 1 \\ &= (16k^2 + 24k + 9) - 1 \\ &= 16k^2 + 24k + 8 \\ &= 8(2k^2 + 3k + 1) \end{aligned}$$

So, $n^2 - 1$ is a multiple of 8. Then, $n(n^2-1)(n+2) = (4k+3)(8(2k^2+3k+1))(4k+3+2)$. Since n^2-1 is a multiple of 8, $n(n^2-1)(n+2)$ is divisible by 8, and hence divisible by 4.

Alternatively, consider the term $n+2$:

$$\begin{aligned} n + 2 &= (4k+3) + 2 \\ &= 4k + 5 \end{aligned}$$

This does not immediately show divisibility by 4. However, we have already shown that n^2-1 is divisible by 8.

Let's also consider the factorization of $n(n^2-1)(n+2)$:

$$n(n^2-1)(n+2) = n(n-1)(n+1)(n+2)$$

This is the product of four consecutive integers. We know that among any four consecutive integers, there must be: 1. One integer divisible by 4. 2. At least one integer divisible by 2.

Let the four consecutive integers be $m, m+1, m+2, m+3$. Their product is $m(m+1)(m+2)(m+3)$.

If m is divisible by 4, then the product is divisible by 4. If $m+1$ is divisible by 4, then the product is divisible by 4. If $m+2$ is divisible by 4, then the product is divisible by 4. If $m+3$ is divisible by 4, then the product is divisible by 4.

This implies that the product of any four consecutive integers is always divisible by 4. In our case, the expression is $n(n-1)(n+1)(n+2)$. Let $m = n-1$. Then the expression is $(m+1)m(m+2)(m+3)$, which is the product of four consecutive integers. Therefore, $n(n^2-1)(n+2)$ is divisible by 4.

Let's reconfirm the original problem statement. The problem statement is "For any integer n, $n(n^2-1)(n+2)$ is divisible by 4." We have shown this by considering the cases of $n \pmod 4$.

Let's use the alternative factorization $n(n-1)(n+1)(n+2)$. The product of four consecutive integers is divisible by $4! = 24$. Thus, it is certainly divisible by 4.

Let's confirm the original expression is equivalent to the product of four consecutive integers. The expression is $n(n^2-1)(n+2)$. $n^2-1 = (n-1)(n+1)$. So, the expression is $n(n-1)(n+1)(n+2)$. This is indeed the product of four consecutive integers: $(n-1), n, (n+1), (n+2)$.

Therefore, since the product of any four consecutive integers is divisible by 4, the expression $n(n^2-1)(n+2)$ is divisible by 4 for any integer n.

Let's provide a proof based on the divisibility of the product of consecutive integers.

We want to prove that $n(n^2-1)(n+2)$ is divisible by 4 for any integer n. We can rewrite the expression as:

$$n(n^2-1)(n+2) = n(n-1)(n+1)(n+2)$$

This is the product of four consecutive integers.

Consider any four consecutive integers: $k, k+1, k+2, k+3$. We will show that their product $k(k+1)(k+2)(k+3)$ is divisible by 4.

Among any four consecutive integers, there must be at least one multiple of 4. Also, among any four consecutive integers, there are at least two even numbers. One of these two even numbers must be a multiple of 4.

Let's be more rigorous. Consider the remainders modulo 4: Case 1: One of the integers is $4m$ for some integer m. The product is divisible by 4. Case 2: None of the integers is a multiple of 4. The remainders modulo 4 could be $1, 2, 3, 0$ (but this is not possible for 4 consecutive). If the sequence starts with $n \equiv 1 \pmod 4$, the integers are $4k+1, 4k+2, 4k+3, 4k+4$. The last integer is divisible by 4. If the sequence starts with $n \equiv 2 \pmod 4$, the integers are $4k+2, 4k+3, 4k+4, 4k+5$. The third integer is divisible by 4. If the sequence starts with $n \equiv 3 \pmod 4$, the integers are $4k+3, 4k+4, 4k+5, 4k+6$. The second integer is divisible by 4.

In all cases, at least one of the four consecutive integers is divisible by 4. Therefore, the product of four consecutive integers is divisible by 4.

Since $n(n^2-1)(n+2) = (n-1)n(n+1)(n+2)$, which is the product of four consecutive integers, it is divisible by 4.

The final answer is $\boxed{n(n^2-1)(n+2) \text{ is divisible by } 4}$.

Problem 66

The problem asks us to prove that a necessary and sufficient condition for a non-negative integer n to be divisible by a positive integer d is that $n \mod d = 0$.

We need to prove two directions: 1. If n is divisible by d, then $n \mod d = 0$. 2. If $n \mod d = 0$, then n is divisible by d.

Part 1: If n is divisible by d, then $n \mod d = 0$.

We are given that n is a non-negative integer and d is a positive integer. We are also given that n is divisible by d.

By the definition of divisibility, if n is divisible by d, then there exists an integer q such that $n = dq$.

The quotient-remainder theorem states that for any integers n and d with $d > 0$, there exist unique integers q and r such that $n = dq + r$ and $0 \leq r < d$.

In our case, we have $n = dq$. We can rewrite this as $n = dq + 0$. Comparing this with the quotient-remainder theorem form $n = dq + r$, we can identify q as the quotient and $r = 0$ as the remainder.

Since $r = 0$, and the condition for the remainder is $0 \leq r < d$, and we are given that d is a positive integer (so $d > 0$), the condition $0 \leq 0 < d$ is satisfied.

The definition of $n \mod d$ is the remainder r when n is divided by d according to the quotient-remainder theorem. Therefore, since $r = 0$, we have $n \mod d = 0$.

This proves the first direction.

Part 2: If $n \mod d = 0$, then n is divisible by d.

We are given that n is a non-negative integer and d is a positive integer. We are also given that $n \mod d = 0$.

By the definition of the modulo operation and the quotient-remainder theorem, $n \mod d = 0$ means that when n is divided by d, the remainder is 0.

According to the quotient-remainder theorem, there exist unique integers q and r such that $n = dq + r$ and $0 \leq r < d$.

Since $n \mod d = 0$, this implies that the remainder r is 0. So, we have $r = 0$.

Substituting $r = 0$ into the equation from the quotient-remainder theorem, we get:

$$n = dq + 0$$

$$n = dq$$

By the definition of divisibility, if there exists an integer q such that $n = dq$, then n is divisible by d. Since we found such an integer q, n is divisible by d.

This proves the second direction.

Since we have proven both directions, we have shown that a necessary and sufficient condition for a non-negative integer n to be divisible by a positive integer d is that $n \mod d = 0$.

Problem 67

Here is the solution to problem 67:

We are asked to prove that for all integers n, $n(n^2-1)(n+2)$ is divisible by 4.

Let $P(n)$ be the statement $n(n^2-1)(n+2)$ is divisible by 4.

We can use the quotient-remainder theorem to divide the integers n into four cases based on their remainder when divided by 4.

Case 1: $n = 4k$ for some integer k.

$$\begin{aligned} n(n^2-1)(n+2) &= (4k)((4k)^2-1)(4k+2) \\ &= (4k)(16k^2-1)(4k+2) \end{aligned}$$

Since one of the factors is $4k$, the entire expression is a multiple of 4. Therefore, $n(n^2-1)(n+2)$ is divisible by 4.

Case 2: $n = 4k+1$ for some integer k.

$$\begin{aligned} n(n^2-1)(n+2) &= (4k+1)((4k+1)^2-1)((4k+1)+2) \\ &= (4k+1)((16k^2+8k+1)-1)(4k+3) \\ &= (4k+1)(16k^2+8k)(4k+3) \\ &= (4k+1)(8k(2k+1))(4k+3) \end{aligned}$$

Since the term $(16k^2+8k)$ has a factor of $8k$, the entire expression is a multiple of 8, and thus a multiple of 4. Therefore, $n(n^2-1)(n+2)$ is divisible by 4.

Case 3: $n = 4k+2$ for some integer k.

$$\begin{aligned} n(n^2-1)(n+2) &= (4k+2)((4k+2)^2-1)((4k+2)+2) \\ &= (4k+2)((16k^2+16k+4)-1)(4k+4) \\ &= (4k+2)(16k^2+16k+3)(4(k+1)) \end{aligned}$$

Since one of the factors is $4(k+1)$, the entire expression is a multiple of 4. Therefore, $n(n^2-1)(n+2)$ is divisible by 4.

Case 4: $n = 4k+3$ for some integer k.

$$\begin{aligned} n(n^2-1)(n+2) &= (4k+3)((4k+3)^2-1)((4k+3)+2) \\ &= (4k+3)((16k^2+24k+9)-1)(4k+5) \\ &= (4k+3)(16k^2+24k+8)(4k+5) \\ &= (4k+3)(8(2k^2+3k+1))(4k+5) \end{aligned}$$

Since the term $(16k^2+24k+8)$ has a factor of 8, the entire expression is a multiple of 8, and thus a multiple of 4. Therefore, $n(n^2-1)(n+2)$ is divisible by 4.

In all four possible cases, $n(n^2-1)(n+2)$ is divisible by 4. Therefore, by the quotient-remainder theorem, the statement $P(n)$ is true for all integers n.

Alternatively, using the factorization: We can factor the expression $n(n^2-1)(n+2)$ further. $n^2-1 = (n-1)(n+1)$. So, $n(n^2-1)(n+2) = n(n-1)(n+1)(n+2)$. This is the product of four consecutive integers.

We can analyze the divisibility by 4 in this product: In any set of four consecutive integers, one of

them must be divisible by 4. Let the four consecutive integers be $m, m+1, m+2, m+3$. If m is divisible by 4, then the product is divisible by 4. If $m+1$ is divisible by 4, then the product is divisible by 4. If $m+2$ is divisible by 4, then the product is divisible by 4. If $m+3$ is divisible by 4, then the product is divisible by 4.

Therefore, the product of any four consecutive integers is always divisible by 4. Since $n(n-1)(n+1)(n+2)$ is the product of four consecutive integers (when arranged in increasing order: $n-1, n, n+1, n+2$ or $n, n+1, n+2, n+3$ if we consider $n+3$), it is always divisible by 4.

The final answer is $\boxed{n(n^2-1)(n+2) \text{ is divisible by } 4}$.

Problem 68

The problem requires solving exercise 68, but there is no exercise numbered 68 in the provided document. The exercises in the document are numbered from 1 to 53. Please provide the correct exercise number you would like solved.

Problem 69

The problem is not numbered 69. I will assume you are referring to question 39 on page 2.

Prove that the sum of any four consecutive integers has the form $4k + 2$ for some integer k.

Let the first integer be n. Since the integers are consecutive, the next three integers are $n + 1$, $n + 2$, and $n + 3$.

The sum of these four consecutive integers is:

$$S = n + (n + 1) + (n + 2) + (n + 3)$$

Combine the terms with n:

$$S = n + n + n + n + 1 + 2 + 3$$

$$S = 4n + 6$$

We want to show that this sum can be written in the form $4k + 2$ for some integer k. We can rewrite the expression $4n + 6$ as follows:

$$S = 4n + 4 + 2$$

Now, we can factor out a 4 from the first two terms:

$$S = 4(n + 1) + 2$$

Let $k = n+1$. Since n is an integer, $n+1$ is also an integer. Therefore, the sum of any four consecutive integers can be written in the form $4k + 2$, where k is an integer.

Thus, the sum of any four consecutive integers has the form $4k + 2$ for some integer k.

Problem 70

The problem asks to evaluate the expressions in Exercise 4.4.70. The prompt states to "Evaluate the expressions in 70." However, the provided document does not contain an Exercise 70. It only contains Exercises up to 40 in the "Evaluate the expressions" section and then continues with other exercises.

Assuming that the intention was to evaluate the expression related to "Test Yourself" section of the document, which is numbered 70, and that this is what is meant by "problem 70", I will proceed with solving that.

The "Test Yourself" section contains the following problems: "1. Integers, n = dq + r; 0 r ¡ d 2. the quotient obtained when n is divided by d; the nonnegative remainder obtained when n is divided by d 3. odd or even 4. 0, 1, 2, ..., (d-1); dq, dq +1, dq +2, ... dq + (d-1) 5. If A, then C; If A, then C; If A, then C 6. If x+y is odd, then x and y are either both odd or both even. 7. If A, then B. If A, then B. If A3, then B. 8. If A, then B. If A, then B. If A3, then B. 9. If A, then B. If A, then B. If A3, then B. 10. The sum of any two integers is odd or even."

It appears that "70" is not a problem number but possibly a typo. However, if "70" refers to a specific problem within the provided text, and given that the context of the document is Elementary Number Theory and Methods of Proof, it is highly probable that "70" is a typo and it refers to one of the numbered problems in the document. Since no specific problem numbered 70 is present, I will assume that the request refers to the *last* problem number listed in the "Evaluate the expressions" section, which is problem 40.

Let's evaluate problem 40 from the document.

Problem 40: For any integer n, $n(n^2-1)(n+2)$ is divisible by 4.

We need to prove that for any integer n, $n(n^2-1)(n+2)$ is divisible by 4. This means that $n(n^2-1)(n+2) \equiv 0 \pmod 4$.

We can analyze this by considering the possible remainders of n when divided by 4.

Case 1: $n \equiv 0 \pmod 4$. If $n \equiv 0 \pmod 4$, then $n = 4k$ for some integer k. Substituting this into the expression:

$$n(n^2-1)(n+2) = (4k)((4k)^2-1)(4k+2)$$

Since there is a factor of $4k$, the entire expression is divisible by 4.

$$n(n^2-1)(n+2) \equiv 0 \cdot ((4k)^2-1)(4k+2) \equiv 0 \pmod 4$$

Case 2: $n \equiv 1 \pmod 4$. If $n \equiv 1 \pmod 4$, then $n = 4k+1$ for some integer k. Substituting this into the expression:

$$n(n^2-1)(n+2) = (4k+1)((4k+1)^2-1)((4k+1)+2)$$

Consider the term (n^2-1):

$$n^2-1 = (4k+1)^2-1 = (16k^2+8k+1)-1 = 16k^2+8k = 8k(2k+1)$$

This term is divisible by 8, and hence by 4. So, $n(n^2-1)(n+2) \equiv (4k+1)\cdot(\text{a multiple of } 8)\cdot(4k+3) \equiv 0 \pmod 4$.

Alternatively, let's look at the factors modulo 4 directly: If $n \equiv 1 \pmod 4$: $n \equiv 1 \pmod 4$ $n^2-1 \equiv 1^2-1 \equiv 1-1 \equiv 0 \pmod 4$ $n+2 \equiv 1+2 \equiv 3 \pmod 4$ So, $n(n^2-1)(n+2) \equiv 1\cdot 0\cdot 3 \equiv 0 \pmod 4$.

Case 3: $n \equiv 2 \pmod 4$. If $n \equiv 2 \pmod 4$, then $n = 4k+2$ for some integer k. Substituting this into the expression:

$$n(n^2-1)(n+2) = (4k+2)((4k+2)^2-1)((4k+2)+2)$$

Consider the term n: $n = 4k + 2 = 2(2k + 1)$. This term is divisible by 2. Consider the term $(n + 2)$: $n + 2 = (4k + 2) + 2 = 4k + 4 = 4(k + 1)$. This term is divisible by 4. Since the term $(n + 2)$ is divisible by 4, the entire product $n(n^2 - 1)(n + 2)$ is divisible by 4.

$$n(n^2 - 1)(n + 2) \equiv n \cdot (n^2 - 1) \cdot 0 \equiv 0 \pmod 4$$

Alternatively, let's look at the factors modulo 4 directly: If $n \equiv 2 \pmod 4$: $n \equiv 2 \pmod 4$ $n^2 - 1 \equiv 2^2 - 1 \equiv 4 - 1 \equiv 3 \pmod 4$ $n + 2 \equiv 2 + 2 \equiv 4 \equiv 0 \pmod 4$ So, $n(n^2 - 1)(n + 2) \equiv 2 \cdot 3 \cdot 0 \equiv 0 \pmod 4$.

Case 4: $n \equiv 3 \pmod 4$. If $n \equiv 3 \pmod 4$, then $n = 4k + 3$ for some integer k. Substituting this into the expression:

$$n(n^2 - 1)(n + 2) = (4k + 3)((4k + 3)^2 - 1)((4k + 3) + 2)$$

Consider the term $(n^2 - 1)$:

$$n^2 - 1 = (4k + 3)^2 - 1 = (16k^2 + 24k + 9) - 1 = 16k^2 + 24k + 8 = 8(2k^2 + 3k + 1)$$

This term is divisible by 8, and hence by 4. So, $n(n^2 - 1)(n + 2) \equiv (4k + 3) \cdot (\text{a multiple of } 8) \cdot (4k + 5) \equiv 0 \pmod 4$.

Alternatively, let's look at the factors modulo 4 directly: If $n \equiv 3 \pmod 4$: $n \equiv 3 \pmod 4$ $n^2 - 1 \equiv 3^2 - 1 \equiv 9 - 1 \equiv 8 \equiv 0 \pmod 4$ $n + 2 \equiv 3 + 2 \equiv 5 \equiv 1 \pmod 4$ So, $n(n^2 - 1)(n + 2) \equiv 3 \cdot 0 \cdot 1 \equiv 0 \pmod 4$.

In all cases, $n(n^2 - 1)(n + 2) \equiv 0 \pmod 4$. Therefore, for any integer n, $n(n^2 - 1)(n + 2)$ is divisible by 4.

Final check of the wording: "Evaluate the expressions in 70." Since problem 70 does not exist, I have proceeded with the assumption it meant problem 40, as it is the last "Evaluate the expressions" problem in the visible page. If "70" referred to something else, the problem would need to be specified.

Let's re-examine if there's any other interpretation for "70". It could be a problem that spans across pages, or a problem number that is implied but not explicitly written. However, given the structure, it's most likely a typo.

If we consider "70" as the *sum* of the last two numbers in the "Evaluate the expressions" section, that would be 39 + 40 = 79, which is not 70.

If "70" is a typo for a different exercise number that actually exists, and if it is from the "Evaluate the expressions" section, the closest numbers to 70 are 39 and 40. Let's analyze problem 39. **Problem 39:** The sum of any four consecutive integers has the form $4k + 2$ for some integer k.

Let the four consecutive integers be n, $n + 1$, $n + 2$, and $n + 3$. Their sum is $S = n + (n + 1) + (n + 2) + (n + 3)$.

First, combine like terms:

$$S = (n + n + n + n) + (0 + 1 + 2 + 3)$$

$$S = 4n + 6$$

Now, we need to show that $4n + 6$ can be written in the form $4k + 2$ for some integer k. We can rewrite $4n + 6$ as follows:

$$4n + 6 = 4n + 4 + 2$$

Factor out 4 from the first two terms:

$$4n + 6 = 4(n + 1) + 2$$

Let $k = n + 1$. Since n is an integer, $n + 1$ is also an integer. Therefore, the sum of any four consecutive integers is of the form $4k + 2$.

This seems like a plausible interpretation if "70" was a typo for 39.

However, if the question implies to evaluate the expression numbered "70" in the "Test Yourself" section, then there is no explicit problem number 70. If "70" refers to the combination of answers for the first 7 items in "Test Yourself", that would be very complex to interpret as "evaluate an expression".

Given the ambiguity of "70", and the fact that "Evaluate the expressions" section has numbers up to 40, the most reasonable interpretation is that "70" is a typo for one of the existing problem numbers. Without further clarification, I will assume it is a typo for the last problem in that section, which is 40. The calculation for problem 40 is shown above.

If "70" is indeed an intended problem, and it is not a typo, and not present in the document, I cannot solve it. However, in an exam scenario, one would ask for clarification.

Assuming the intent was problem 40, the steps are shown. Let's consider if "70" refers to a combination of problems. For example, if it implies the sum of the results of problems 1 and 69 (if 69 existed). This is also too speculative.

Let's assume "70" refers to a *concept* related to the number 70. For example, the properties of the number 70 itself. But this is not an "expression to evaluate".

Given the problem statement format, "Evaluate the expressions in 70," and the context of exercises, it is overwhelmingly likely to be a problem number. The absence of problem 70 necessitates an assumption. The most logical assumption, when faced with a missing numbered item in a sequential list, is that it's a typo for the next item or a nearby item.

Since the "Evaluate the expressions" section ends at 40, and if we consider the possibility of a typo for a problem that *should* exist but is missing, that would be highly speculative.

Let's assume "70" is a typo for a problem from the "Test Yourself" section that involves an expression to evaluate. The items in "Test Yourself" are definitions or proofs. However, item 6 in "Test Yourself" states: "If x+y is odd, then x and y are either both odd or both even." This is a statement to prove, not an expression to evaluate.

Therefore, the most robust approach is to assume "70" is a typo for one of the explicitly numbered problems in the "Evaluate the expressions" section. Since problem 40 is the last one in that list, and it fits the format of "evaluating an expression" by proving a divisibility property, I will stand by the solution for problem 40.

The final answer is $\boxed{n(n^2 - 1)(n + 2) \text{ is divisible by } 4}$.

Problem 71

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We are asked to evaluate the expression 43 div 9. This is equivalent to finding the quotient when 43 is divided by 9.

We want to find integers q and r such that $43 = 9q + r$ and $0 \leq r < 9$.

We can perform the division: $43 \div 9$.

We can see how many times 9 fits into 43. $9 \times 1 = 9$ $9 \times 2 = 18$ $9 \times 3 = 27$ $9 \times 4 = 36$ $9 \times 5 = 45$

The largest multiple of 9 that is less than or equal to 43 is 36, which is 9×4. So, the quotient q is 4.

Now we find the remainder r by subtracting the product of the quotient and the divisor from the dividend: $r = 43 - (9 \times 4)$

$$r = 43 - 36$$
$$r = 7$$

We check if the remainder r satisfies the condition $0 \leq r < 9$. In this case, $r = 7$, and $0 \leq 7 < 9$, which is true.

Therefore, 43 div $9 = 4$.

The final answer is $\boxed{4}$.